Public-Matrix Symmetry Yields Faster Forgery Attacks on Odd-Characteristic SNOVA

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Abstract

We show that the odd-characteristic ($q = 19$) parameter sets of SNOVA Version 2.3 admit faster forgery attacks than were previously known, because the public design represents many of its quadratic features twice. Each feature is labeled by an *ordered* pair of powers from the public algebra $A = \mathbb{F}_q[S]$, but symmetry of the public base matrices leaves the polynomial unchanged when the two labels are reversed. Thus, if $d = \dim_{\mathbb{F}_q} A$, the d^2 labeled power-pair features of each public form collapse to at most $\binom{d+1}{2}$ distinct ones—16 to 10 at block size $\ell = 4$, and 4 to 3 at $\ell = 2$. This identification occurs before the public outer map and holds for every symmetric public key; it is not a weak-key event.

Combining this symmetry quotient with exact affine elimination, we construct for each of the nine public $q = 19$ parameter sets a publicly checkable reduction to a smaller forgery system that drops no verifier equations. Under the stated global-rank conditions and the cited MQ cost models, the modeled expected operation counts at NIST Levels I, III, and V are, respectively, $2^{128.82}$–$2^{129.44}$, $2^{171.69}$–$2^{183.44}$, and $2^{214.12}$–$2^{235.19}$. Recosting the field arithmetic charged by those models as validated two-input AND/XOR/XNOR circuits gives $2^{132.17}$–$2^{133.75}$, $2^{175.04}$–$2^{187.74}$, and $2^{217.48}$–$2^{239.50}$. These figures price dominant arithmetic rather than a complete attack: the global rank spectra and production solver behavior remain the principal open inputs.

1 Introduction

The cryptanalytic task. A multivariate signature scheme publishes a polynomial map of total degree at most two,

$$\mathcal{P} : \mathbb{F}_q^N \longrightarrow \mathbb{F}_q^M.$$

For a message with hash-derived target $t \in \mathbb{F}_q^M$, the algebraic core of a forgery is to find x satisfying $\mathcal{P}(x) = t$; any additional format or rejection checks must also pass. This is an MQ problem: M equations of degree at most two in the coordinates of x. Linear and constant terms are allowed; throughout the paper, "quadratic" refers to the maximum degree. The apparent equation and variable counts are only a first approximation to security. Public algebraic structure can make different-looking quadratic features identical, can expose linear equations, or can split the remaining variables into blocks that a specialized solver can exploit.

SNOVA is one of the nine digital-signature candidates that NIST advanced to the third round of its additional post-quantum signature process in May 2026 [14]. It stores the signature variables in

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small matrices and builds its public quadratics from pairs of powers of a public matrix S. Beullens showed that an attacker can impose an affine relation among the signature columns and thereby reduce forgery to a smaller underdetermined MQ system [2]. We find an additional, deterministic reduction inside the quadratic features themselves.

The hidden duplication. Let P_i be a public symmetric base matrix, let $A = \mathbb{F}_q[S]$, and write $\Sigma_\xi = I_n \otimes \xi$ for $\xi \in A$. The feature with left label ξ and right label η is

$$q_{i,\xi,\eta}(u) = u^\mathsf{T} \Sigma_\xi P_i \Sigma_\eta u.$$

Because the matrices are symmetric and the result is a scalar,

$$q_{i,\xi,\eta}(u) = q_{i,\eta,\xi}(u). \tag{1}$$

The verifier begins from ordered labels, but the polynomial depends only on the corresponding unordered pair. We call the operation of identifying these duplicate labels the *symmetry quotient*.

For a two-dimensional toy algebra with basis $\{1, S\}$, the four labels

$$(1,1), \qquad (1,S), \qquad (S,1), \qquad (S,S)$$

produce only three polynomials because the middle two are equal. More generally, if $\dim_{\mathbb{F}_q} A = d$, then at most

$$K = m_1 \binom{d+1}{2} \tag{2}$$

independent power-pair features remain across the m_1 base forms, rather than $m_1 d^2$. Thus each $\ell = 4$ base form drops from 16 ordered labels to 10 unordered labels; each $\ell = 2$ base form drops from 4 to 3. The loss occurs before the public outer linear map—the final public mixing of these features into verifier outputs—so later randomization cannot recreate the discarded directions. It holds for every symmetric public key and every attacker-chosen column relation.

Core idea. The verifier assigns two names to the same quadratic polynomial whenever it reverses a pair of power labels. The attack first removes those duplicate features, then uses ordinary linear algebra to retain all remaining verifier equations, and finally solves only the smaller quadratic core.

The attack as a data-flow. The proof is easiest to follow as a sequence of exact transformations. The word *lift* below means the attacker-chosen affine rule that turns one column vector into a full signature candidate.

Full public verifier	$\longrightarrow$	impose one affine common-column relation; quadratic, linear, and constant terms become explicit
	$\downarrow$	
Symmetry quotient	$\longrightarrow$	identify (ξ, η) with (η, ξ) and keep only the distinct quadratic features
	$\downarrow$	
Exact affine elimination	$\longrightarrow$	use a left-kernel projection to expose every output direction unreachable by the quadratics as an ordinary linear constraint
	$\downarrow$	
Structured residual MQ	$\longrightarrow$	choose a stable four-block slice for $\ell = 4$, or a rejection-compatible relation for $\ell = 2$
	$\downarrow$	
Candidate signature	$\longrightarrow$	map back to the original coordinates and run the unmodified public verifier

Every transformation through the residual MQ system is exact and uses only public algebra and linear algebra. The solver model enters only when we estimate the cost of solving that residual system. The final public-verifier check makes the attack Las Vegas: a trial may be discarded, but an invalid signature is never returned.

Four jobs that a complete attack must perform. The equality (1) is only the first job. The construction also has to preserve all public outputs, ensure that targets have roots, and make the residual equations compatible with a practical solver:

1. **Constrain the signature columns.** Every signature column is written as an affine function of one common vector. Substitution separates the verifier into quotient quadratic features, offset-linear terms, and constants.
2. **Eliminate the affine part without dropping equations.** Output directions outside the quadratic image become linear constraints. For each target satisfying those affine consistency conditions, Gaussian elimination leaves a smaller MQ system with exactly the same solutions as the constrained verifier.
3. **Choose a solver-friendly search space.** For $\ell = 4$, a structured choice of offsets leaves a large A-linear kernel. Over a splitting field this kernel becomes four variable blocks, preserving the multidegrees used by special XL. For $\ell = 2$, fixed-skew and filtered relations handle the verifier's rejection rule directly.
4. **Retry and verify.** Exact moment bounds control the probabilities of root existence and uniqueness for random target–slice pairs. A candidate recovered by the MQ solver is transformed back and checked by the original verifier.

A complete dimension walk-through. The Level-I parameter set $(v, o, \ell, r) = (28, 5, 4, 4)$ illustrates where the reduction comes from: it has 28 vinegar blocks, 5 oil blocks, and blocks with 4 rows and 4 columns. Here $n = v + o = 33$, one stacked signature column has $N_0 = n\ell = 132$ coordinates, the verifier has $M = or\ell = 80$ scalar outputs, and there are $m_1 = 5$ public base-form groups.

Table 1: The dimensions in the $(28, 5, 4, 4)$ attack. This table is only a worked example; the general formulas appear in later sections.

Stage	Dimension	Meaning
Raw public feature vector	$N_{\text{raw}} = 1{,}280$	$m_1\ell^2r^2$ labeled slots before the column relation
After the common-column relation	80 ordered quadratics	one feature for each of $m_1\ell^2 = 5 \cdot 16$ ordered power pairs
After the symmetry quotient	$K = 50$ distinct quadratics	$m_1\binom{\ell+1}{2} = 5 \cdot 10$ unordered power pairs
Affine output directions	$M - K = 30$	linear equations that determine an affine search space
Full residual domain	102 coordinates	$N_0 - (M - K) = 132 - 30$ after affine elimination
Stable kernel	52 base-field dimensions	an A-linear space of A-dimension 13 on which offset-linear motion vanishes
Selected special-XL slice	$h = 40$	four blocks of $s = 10$ variables; hence $\tau = K - h = 10$

The final slice deliberately has fewer variables than equations. Under full cross-polar rank, a random target–slice trial then contains a unique base-field root with probability essentially 19^{-10}. Paying roughly 19^{10} retries is worthwhile because reducing the Macaulay matrix from the full residual domain to four blocks of ten variables saves much more. This is why the paper analyzes both the algebraic reduction and the root-retry probability, rather than quoting an MQ dimension alone.

Headline estimates. Table 2 reports modeled costs and the differences between them. "Model" is the base-2 expected-cost exponent in the *pinned* attack model, meaning that we fix the cited estimator, its version, and its conventions exactly as specified in the artifact. It starts from the cited per-solve MQ operation formula and, when the solve is repeated, includes the explicit target–slice retry factor. Target generation performed before a single solve is accounted for separately. "AXN arith." changes only the unit for the arithmetic charged by that model, replacing its coarse field-operation factor with validated signed two-input AND/XOR/XNOR (AXN) circuits. "Gain over generic" is the reduction in the Model exponent relative to a pinned generic-MQ baseline on the same symmetry quotient. It is not always a solver improvement: for the structured rows it comes from the stable four-block slice and special XL, whereas for the filtered Level-V row it comes from the zero-offset filtered lift. "Arithmetic budget" is the nominal NIST reference minus the AXN arithmetic exponent.

Table 2: Headline conditional estimates for the public Version 2.3 parameter sets. The tuple is (v, o, ℓ, r); its meaning is given in Section 3.1. Model and AXN are base-2 cost exponents; gain over generic and arithmetic budget are differences of exponents, measured in bits. V2.3 reproduces the official "log #gates" table.

Level	Parameters	Mode	V2.3	Model	AXN arith.	gain over generic	Arith. budget
I	$(28, 5, 4, 4)$	structured XL	194	128.82	132.17	7.22	10.83
I	$(48, 16, 2, 2)$	fixed-skew MQ	164	129.44	133.75	0.00	9.25
I	$(28, 4, 4, 5)$	structured XL	194	128.82	132.17	7.22	10.83
III	$(40, 7, 4, 4)$	structured XL	264	171.69	175.04	8.91	31.96
III	$(72, 24, 2, 2)$	fixed-skew MQ	234	183.44	187.74	0.00	19.26
III	$(38, 5, 4, 5)$	structured XL	239	171.69	175.04	8.91	31.96
V	$(50, 9, 4, 4)$	structured XL	333	214.12	217.48	10.56	54.52
V	$(96, 32, 2, 2)$	filtered MQ	303	235.19	239.50	2.65	32.50
V	$(52, 6, 4, 6)$	structured XL	333	214.12	217.48	10.56	54.52

For example, the first row's AXN entry 132.17 means about $2^{132.17}$ charged two-input arithmetic gates under the special-XL model and its unique-root retry factor.

Conditions behind the table. The structured-XL rows require two separate production assumptions: every nonzero output direction must have full cross-polar rank on the chosen slice, and a planted-solvable special-XL Macaulay matrix (one generated together with a known root) must have the predicted one-dimensional right kernel. The fixed-skew Level-I and Level-III rows use the sufficient all-target root conditions $r_* \geq 96$ and $r_* \geq 144$. The filtered Level-V row uses the accepted-root condition of Proposition 7, implied there by $r_* \geq 240$. Each r_* condition is global: it concerns every nonzero output combination, not only the sampled directions in the artifact. Lower ranks are handled by explicit retry bounds in Section 6.

The official Version 2.3 values sit 30.25–115.52 bits above our AXN arithmetic column, but the two use different circuit methodologies, so the gap is not a clean measure of our improvement. For a like-for-like comparison, we recost the earlier ordered-label affine-column estimator in the same AXN basis; the per-row reductions are then 38.02–129.66 bits (Table 12). The ordered-label system is executable, but by Theorem 2 its reversed labels are not independent features on a symmetric key, so feeding the ordered-label count to a generic estimator overcounts the true quadratic space.

NIST describes its category references across the security metrics it regards as relevant, rather than as a gate-count-only test [15]. We therefore use 143/207/272 only as nominal arithmetic reference points.

How to read the paper. Section 2 fixes the background and notation. The exact attack is in Sections 3 to 5. Root existence and retry probabilities are separated in Section 6; the solver and gate models are isolated in Section 7; and Section 8 states exactly which assumptions are tested rather than proved. All long proofs and exhaustive cost-search certificates are deferred to the appendices.

The submitted $q = 16$ matrices are not symmetrized and therefore do not satisfy the hypothesis of our quotient theorem. Returning unchanged to that design, however, would restore the known characteristic-two attack surface discussed in Section 2.4.

2 Mathematical Background and Relation to Prior Work

This section fixes the notation and background used throughout: underdetermined MQ, polar forms, Macaulay matrices, and multigraded XL.

2.1 MQ systems, roots, and affine slices

Write $\mathbb{F}_q$ for the finite field with q elements. An MQ system over $\mathbb{F}_q$ is a vector-valued polynomial map of total degree at most two,

$$g = (g_1, \ldots, g_K) : \mathbb{F}_q^N \longrightarrow \mathbb{F}_q^K,$$

together with a target $z \in \mathbb{F}_q^K$. We continue to call g quadratic when some coordinates also contain linear or constant terms. A *root* is an x with $g(x) = z$. The set

$$g^{-1}(z) = \{x : g(x) = z\}$$

is the *fiber* above z: it is the set of all candidate assignments that produce the target. After the affine reconstruction described later, those assignments become full signature candidates. All arithmetic is over the stated finite field; every concrete SNOVA calculation in this paper uses $q = 19$, so additions and multiplications are performed modulo 19.

As a toy example, two equations such as

$$x_1x_2 + x_3^2 = z_1, \qquad x_1^2 + x_2x_3 + x_1 = z_2$$

form an MQ system in three variables. Having more variables than equations makes the system *underdetermined*, but does not make it linear or necessarily easy. Modern MQ attacks often spend most of their time on very large linear-algebra matrices built from the quadratic equations.

An *affine space* is a translate $a + V = \{a + v : v \in V\}$ of a linear subspace. An *affine slice* $a + L \subseteq a + V$ restricts the search to a smaller direction space L. If $h = \dim L$, choosing a basis of L turns the slice into an MQ system in h coordinates. Smaller h makes the algebraic solve cheaper, but it also lowers the chance that the slice contains a root; Section 6 quantifies this tradeoff.

Three logically distinct events recur throughout the paper:

(i) the chosen target and slice contain at least one $\mathbb{F}_q$-rational root;

(ii) the slice contains exactly one such root; and

(iii) the particular Macaulay matrix used by special XL has a one-dimensional right kernel from which row reduction reveals that root.

The first two are properties of the quadratic map. The third is a statement about a specific solver representation. A proof of root existence is not, by itself, a proof that one solver invocation recovers the root.

2.2 Rank, polar forms, and collisions

For a scalar quadratic polynomial f, define its polar form by

$$\beta_f(x, y) = f(x + y) - f(x) - f(y) + f(0).$$

In odd characteristic, β_f is bilinear. It is the finite-field analogue of the part of a directional derivative that comes from the quadratic terms. Its matrix rank measures how many independent directions the quadratic part couples.

For a vector-valued map $g = (g_1, \ldots, g_K)$, every nonzero coefficient vector $b \in \mathbb{F}_q^K$ selects a scalar combination $b \cdot g$. The minimum polar rank must be taken over *all* such output directions. Multiplying b by a nonzero scalar does not change the rank, so these vectors can be grouped into projective directions: one-dimensional lines through the origin.

Rank matters because finite-field Fourier (character-sum) identities imply that a scalar quadratic with high polar rank is well balanced. Applied to all nonzero combinations $b \cdot g$, this bounds every target fiber rigorously: when the ranks are high enough relative to K, each fiber has size close to the random-map prediction q^{N-K}. A related *cross-polar rank* measures how strongly a displacement inside a slice interacts with the full residual domain; high cross-polar rank suppresses pairs of distinct slice points that map to the same target. In summary:

Polar rank	controls whether full-space target fibers are nonempty and close to their expected size.
Cross-polar rank	controls collisions inside an affine slice and therefore unique-root retry probabilities.
Macaulay kernel	its dimension controls whether the chosen special-XL linear algebra actually exposes the root.

High rank is not invoked as a vague "randomness" heuristic in the structural proofs: Section 6 gives exact first- and second-moment formulas. The production issue is whether the required rank lower bound holds in every nonzero output direction. Public sampling gives evidence, but only a proof or exhaustive enumeration certifies the global minimum.

2.3 XL and Macaulay matrices

XL ("eXtended Linearization") turns a nonlinear system into a larger linear system. Starting from quadratic equations $f_i = 0$, it multiplies each f_i by selected monomials. For example,

$$f = x_1x_2 + x_3^2 + x_1 + 1 \quad \Longrightarrow \quad x_1 f = x_1^2x_2 + x_1x_3^2 + x_1^2 + x_1.$$

Each resulting polynomial becomes a row. Each monomial—for example $x_1^2x_2$ or $x_1x_3^2$—becomes a column. The coefficient matrix is a *Macaulay matrix*. Gaussian elimination on this matrix can reveal enough relations among the monomials to recover a root. Treating monomials as columns is only a linear-algebra device: the monomials are not independent variables in the original problem, and the solver must eventually recover coordinates that satisfy all of their multiplicative relations.

The cost comes from the size and rank of this matrix, not from evaluating the original equations once. Generic XL tracks only total degree. *Special XL* exploits a variable-block structure. A monomial then has a vector of degrees, one entry per block, called its *multidegree*. Building only selected multidegrees can make the Macaulay matrix much smaller.

Our $\ell = 4$ reduction produces four blocks after a change to eigenblock coordinates. A quadratic using one block twice has degree pattern $(2, 0, 0, 0)$ up to permutation; a bilinear equation using two blocks has pattern $(1, 1, 0, 0)$. The stable-lift construction in Section 5 exists precisely to preserve these patterns after affine elimination. A *homogenizer* is one extra variable per block that allows the constant and linear terms to be written with the same multidegree; it is set to one when the root is read out.

If x is a root, the vector of all selected monomial values

$$\nu(x) = (1, x_1, \ldots, x_N, x_1^2, x_1x_2, \ldots)$$

lies in the right kernel of the dehomogenized Macaulay matrix. The right kernel is the space of column vectors w with $\mathcal{M}w = 0$. If it is one-dimensional, normalizing its constant coordinate to one reveals the degree-one coordinates and hence the root. A unique base-field root does not automatically imply a one-dimensional Macaulay kernel: extension-field roots or algebraic relations among rows can create extra kernel vectors. This is the separate production solver assumption in the structured attacks.

2.4 Relation to prior attacks

Beullens rewrote the SNOVA verifier in ordered bilinear power-pair coordinates and derived affine-column forgery attacks from rank defects in the resulting outer matrix [2]. Our quotient acts one layer earlier: public-matrix symmetry identifies the power-pair polynomials *before* the outer map sees them.

Cabarcas et al. exploit stability under the public matrix algebra with a multigraded XL algorithm [4]. Furue et al. expose related multihomogeneous systems through matrix transformations [5]. Our attack starts from the complete affine forgery system, uses a public column lift to retain every output equation, and then restricts to an invariant kernel on which the affine offsets preserve the block grading needed by special XL.

The wedge attack breaks the submitted characteristic-two parameters through the block-ring structure [3]; later work studies the wedge map further [19]. Other analyses reinterpret scalar-expanded SNOVA as Unbalanced Oil and Vinegar (UOV), or lift it to extension-field UOV, for key recovery [7, 10, 12]. Symmetric and exterior algebras also appear in recent UOV key-recovery attacks [8, 16]. Our attack is a direct forgery attack: it does not recover the hidden oil subspace that constitutes the UOV-style secret key.

For the generic $\ell = 2$ cores, we reproduce Hashimoto's optimization, including its published boundary value $a = 1$ [6, 2]. When this leaves the elementary system $\mathrm{MQ}(q, 1, 1)$, the artifact solves it by exhaustive evaluation and charges that work conservatively. May, Ostuzzi, and Ressler's *Just Guess* algorithm and the generalized variable partitioning of Asanuma et al. are more recent approaches to underdetermined MQ [11, 1]. Evaluating the released classical Just-Guess formulas on our residual dimensions gives larger estimates than the selected Beullens–Hashimoto modes. We have not implemented the newer generalized partition search. The "generic MQ" numbers below are therefore pinned, reproducible baselines, not a claim to have exhausted every generic solver.

Solver engineering is also moving quickly. Sakata and Takagi use Hilbert-series information to reduce the matrices built by an F_4 Gröbner-basis solver and report a new MQ-challenge record [17]. That is an implementation and challenge result, not a plug-in cost formula for the underdetermined residual systems here, so we do not substitute it into Table 2.

2.5 Notation at a glance

Table 3: Recurring notation. Generic background notation such as N is local; N_0 is the specific common-column dimension in the SNOVA reduction.

Symbol	Meaning
(v, o, ℓ, r)	Public parameter tuple: v vinegar blocks, o oil blocks, block height ℓ, and r columns per signature block.
$n = v + o$	Number of matrix-valued signature blocks.
$M = or\ell$	Number of scalar public output equations.
N	Generic domain dimension for an MQ map $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$; its meaning is local to the section in which the map is defined.
$N_0 = n\ell$	Length of one stacked signature column after the common-column relation.
$N_{\mathrm{raw}} = m_1\ell^2 r^2$	Number of labeled verifier feature slots before imposing the relation.
$A = \mathbb{F}_q[S]$	Public commutative matrix algebra generated by S; in the Version 2.3 rows, $\dim_{\mathbb{F}_q} A = \ell$.
m_1	Number of public base-form groups in the pinned implementation.
$K = m_1\binom{\ell+1}{2}$	Maximum number of distinct quadratic power-pair features after the symmetry quotient.
Q_R	Matrix mapping the K quotient features to public output directions.
$C_{R,\mathbf{v}}$	Linear constraint matrix obtained by projecting away the quadratic image.
$H_{R,\mathbf{v}}$	Stable A-linear kernel used to choose structured solver slices.
L, h	Slice direction space and its base-field dimension.
$\tau = K - h$	Slice deficit. It enters the exact full-rank retry formula; for the selected $\tau = 10, 14$ slices, the retry count is essentially q^τ.
r_*	Minimum full-space polar rank over all nonzero scalar output combinations.
$e_L(b)$	Cross-polar rank of output direction b relative to slice L.
μ, $\bar{\eta}$	Expected roots on a random target–slice pair and an upper bound on its collision mass.

3 The Affine-Column Framework

This section rewrites the public verifier after an attacker-imposed affine relation among signature columns. The result has three visibly separate parts: quotient quadratic features, linear terms created by the offsets, and a constant term. Proposition 1 then eliminates the affine part exactly.

Three vector spaces appear and should not be confused. The raw public feature vector has length N_{raw}; the verifier outputs M scalar equations; and the common column variable u has length N_0. The matrices below simply move between these three spaces.

3.1 Parameters and public forms

The public parameter tuple is (v, o, ℓ, r). There are v vinegar blocks and o oil blocks, for a total of

$$n = v + o$$

matrix-valued signature blocks. The names "vinegar" and "oil" come from SNOVA's UOV ancestry; in this public forgery attack they serve only to specify dimensions, and the attacker never learns or reconstructs the secret partition. Each block has ℓ rows and r columns. Stacking the jth column of all n blocks gives a vector of length $N_0 = n\ell$; the verifier has $M = or\ell$ scalar outputs. The pinned Version 2.3 implementation defines

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \qquad M = or\ell, \qquad N_0 = n\ell. \tag{3}$$

Here m_1 is the number of public base-form groups used to assemble the outputs. The ceiling is used by the reference implementation; a public analysis script uses a floor and therefore disagrees when $\ell \nmid or$. All dimensions in this paper follow the pinned implementation [18].

Let $S \in \mathrm{Mat}_\ell(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial. Its powers generate the commutative matrix algebra

$$A = \mathbb{F}_q[S], \qquad \Sigma_\xi = I_n \otimes \xi \quad (\xi \in A).$$

For the Version 2.3 parameters, $A \cong \mathbb{F}_{q^\ell}$ and $\dim_{\mathbb{F}_q} A = \ell$. Multiplication by Σ_ξ applies the same $\ell \times \ell$ matrix ξ independently to every signature block.

The scalar-expanded public base matrices are $P_1, \dots, P_{m_1} \in \mathrm{Mat}_{N_0}(\mathbb{F}_q)$. In the public $q = 19$ construction, each P_i is symmetric. For $\xi, \eta \in A$, define the bilinear form

$$B_i^{\xi,\eta}(x, y) = x^\mathsf{T} \Sigma_\xi P_i \Sigma_\eta y. \tag{4}$$

Setting $x = y$ makes this a quadratic feature. The power basis $1, S, \dots, S^{\ell-1}$ gives the ordered power-pair coordinates used in the prior affine-column attack.

3.2 Imposing one common column variable

Write the stacked signature columns as

$$U = [u_0 | \cdots | u_{r-1}], \qquad u_j \in \mathbb{F}_q^{N_0}.$$

The attacker chooses multipliers $R_j \in A$ and offsets $v_j \in \mathbb{F}_q^{N_0}$, with $R_0 = 1$. Write $R = (R_0, \dots, R_{r-1})$ and $\mathbf{v} = (v_0, \dots, v_{r-1})$, and impose

$$u_j = \Sigma_{R_j} u + v_j, \qquad 0 \le j < r. \tag{5}$$

Thus all r columns are affine functions of one vector u. This is not an assumption about honest signatures; it defines a publicly checkable subset in which the attacker searches for a forgery.

Before the relation is imposed, the feature vector has

$$N_{\mathrm{raw}} = m_1 \ell^2 r^2$$

coordinates: m_1 base forms, ℓ^2 ordered power pairs, and r^2 ordered column pairs. The official verifier flattens each $r \times \ell$ block in (i, j) order, whereas our feature formulas use the block-transposed (j, i) order. Let $t_{\mathrm{off}} \in \mathbb{F}_q^M$ and $\mathcal{E}_{\mathrm{off}} \in \mathbb{F}_q^{M \times N_{\mathrm{raw}}}$ denote the target and public outer map in the official order. If Π_{out} is the corresponding public permutation, define

$$t = \Pi_{\mathrm{out}} t_{\mathrm{off}}, \qquad \mathcal{E} = \Pi_{\mathrm{out}} \mathcal{E}_{\mathrm{off}}.$$

This is only a coordinate reordering; it changes neither ranks nor solutions.

After substituting (5), products of two variable parts give quadratic terms in u, products of one variable part and one offset give linear terms, and products of two offsets give constants. For a fixed relation $R = (R_j)_{j=0}^{r-1}$, let $\mathcal{R}_R \in \mathbb{F}_q^{N_{\mathrm{raw}} \times m_1 \ell^2}$ expand the remaining ordered quadratic features into the full feature vector, and let $\mathcal{F}_{R,\mathbf{v}} \in \mathbb{F}_q^{N_{\mathrm{raw}} \times N_0}$ collect the offset-linear coefficients. The complete substituted verifier is

$$\mathcal{E}(\mathcal{R}_R z_{\mathrm{ord}}(u) + \mathcal{F}_{R,\mathbf{v}} u) + c_{R,\mathbf{v}} = t, \tag{6}$$

where $z_{\mathrm{ord}}(u)$ contains the $m_1 \ell^2$ values $B_i^{S^a,S^b}(u,u)$ and $c_{R,\mathbf{v}}$ is the public constant term. Put

$$E_R = \mathcal{E}\mathcal{R}_R \in \mathbb{F}_q^{M \times m_1 \ell^2}, \qquad \Lambda_{R,\mathbf{v}} = \mathcal{E}\mathcal{F}_{R,\mathbf{v}} \in \mathbb{F}_q^{M \times N_0}. \tag{7}$$

3.3 Separating quadratic and affine output directions

By Theorem 2, the ordered feature vector duplicates every off-diagonal power pair. Let $z_{\mathrm{sym}}(u)$ contain one coordinate for each unordered pair $0 \le a \le b < \ell$, and let D copy each off-diagonal coordinate into its two ordered positions. Then $z_{\mathrm{ord}}(u) = D z_{\mathrm{sym}}(u)$. Define

$$Q_R = E_R D \in \mathbb{F}_q^{M \times K}, \qquad K = m_1 \binom{\ell+1}{2}. \tag{8}$$

The columns of Q_R span all output directions that the residual quadratic features can reach.

Now choose W_R whose rows form a basis of the left kernel of Q_R and put

$$C_{R,\mathbf{v}} = W_R \Lambda_{R,\mathbf{v}}. \tag{9}$$

Multiplying by W_R kills every quadratic feature. What remains is therefore an ordinary affine system in u. This elementary projection is the key to retaining the *complete* verifier rather than dropping output equations that do not lie in the quadratic image.

Q_R	sends the K distinct quadratic features to the M public output coordinates; its column space is the quadratic image.
W_R	keeps precisely the output directions orthogonal to that image, so $W_R Q_R = 0$.
$C_{R,\mathbf{v}} = W_R \Lambda_{R,\mathbf{v}}$	records the ordinary linear equations that remain in those output directions after the offsets are introduced.

A two-output example. Suppose that, after the column substitution, two verifier outputs over $\mathbb{F}_{19}$ are

$$y_1 = q(u) + \ell_1(u), \qquad y_2 = 2q(u) + \ell_2(u),$$

where q is quadratic and ℓ_1, ℓ_2 are affine. The combination $2y_1 - y_2 = 2\ell_1(u) - \ell_2(u)$ cancels the quadratic term. Solving this affine equation and substituting it back leaves one quadratic equation; neither original output was dropped. The rows of W_R perform the same cancellation simultaneously for the complete verifier.

Proposition 1 (Exact affine reduction)**.** *Let $\rho = \operatorname{rank} Q_R$, $\lambda = \operatorname{rank} C_{R,\mathbf{v}}$, and $\delta = M - \rho - \lambda$. Then*

$$\operatorname{rank}[Q_R \ \Lambda_{R,\mathbf{v}}] = \rho + \lambda,$$

where $[Q_R \ \Lambda_{R,\mathbf{v}}]$ denotes horizontal matrix concatenation. After row reduction, exactly δ independent affine consistency conditions remain on the target. For every target satisfying those conditions, Gaussian elimination leaves ρ equations of degree at most two in $N_0 - \lambda$ variables, with exactly the same solution set as (6).

In particular, if $\rho = K$ and $\lambda = M - K$, every target passes the affine consistency test*, and the residual system has K equations of degree at most two on an affine translate of* $\ker C_{R,\mathbf{v}}$*. This conclusion does not assert that the residual system has a root.*

Proof. The row space of W_R is $\ker Q_R^{\mathsf{T}}$. Since W_R has full row rank,

$$\ker W_R = \left(\ker Q_R^{\mathsf{T}}\right)^{\perp} = \operatorname{im} Q_R.$$

Thus W_R realizes the quotient map $\mathbb{F}_q^M \to \mathbb{F}_q^M / \operatorname{im} Q_R$. The image of the columns of $\Lambda_{R,\mathbf{v}}$ in this quotient has dimension $\operatorname{rank}(W_R \Lambda_{R,\mathbf{v}}) = \lambda$, which proves $\operatorname{rank}[Q_R \ \Lambda_{R,\mathbf{v}}] = \rho + \lambda$.

Write $b = t - c_{R,\mathbf{v}}$. Projecting the substituted verifier by W_R gives

$$C_{R,\mathbf{v}} u = W_R b.$$

The codomain of $C_{R,\mathbf{v}}$ has dimension $M - \rho$, while its image has dimension λ. Solvability therefore imposes exactly $(M - \rho) - \lambda = \delta$ independent affine conditions on the target. For a target satisfying the affine consistency conditions, parameterize the solution space of these affine equations by $N_0 - \lambda$ free variables and substitute into a complementary set of ρ rows. This leaves ρ equations of degree at most two and uses only invertible row operations and exact affine elimination, so the solution set is unchanged. If $\rho = K$ and $\lambda = M - K$, then $\delta = 0$ and $C_{R,\mathbf{v}}$ is surjective onto its $M - K$-dimensional codomain; hence every target passes the affine consistency test. □

Operational meaning. The attacker computes Q_R, W_R, and $C_{R,\mathbf{v}}$ from the public key before running an MQ solver. If the ranks are unfavorable, the relation or offsets are changed. When $\operatorname{rank} Q_R = K$ and $\operatorname{rank} C_{R,\mathbf{v}} = M - K$, every hash target determines a nonempty affine space $a(t) + \ker C_{R,\mathbf{v}}$ on which K equations of degree at most two remain. This is an exact reformulation that discards no verifier equation; whether the residual equations actually have a root is analyzed in Section 6. Conversely, every root of the residual system maps back through the column relation to a solution of the complete substituted verifier.

4 The Symmetry Quotient

The central reduction is the formal version of (1). Readers may interpret $\operatorname{Sym}^2(A)$ simply as the vector space spanned by *unordered* pairs from A; the tensor notation records why the identification is basis-independent.

For a fixed base form P_i, define

$$q_{i,a,b}(u) = u^{\mathsf{T}} \Sigma_{S^a} P_i \Sigma_{S^b} u.$$

Transposition gives $q_{i,a,b} = q_{i,b,a}$. In coordinates, the feature vector therefore contains two copies of every off-diagonal polynomial. The theorem below shows that this is not an accident of the power basis: every alternating label $\xi \otimes \eta - \eta \otimes \xi$ vanishes. Write $\mathbb{F}_q[u]_2^{\mathrm{hom}}$ for the vector space of homogeneous quadratic polynomials in the coordinates of u.

Theorem 2 (Symmetric-square factorization)**.** *Let $A = \mathbb{F}_q[S]$ have dimension d, and assume $S^{\mathsf{T}} = S$ and $P_i^{\mathsf{T}} = P_i$. For each i, the map*

$$A \otimes_{\mathbb{F}_q} A \longrightarrow \mathbb{F}_q[u]_2^{\mathrm{hom}}, \qquad \xi \otimes \eta \longmapsto u^{\mathsf{T}} \Sigma_\xi P_i \Sigma_\eta u$$

is invariant under $\xi \otimes \eta \mapsto \eta \otimes \xi$. It therefore factors through

$$\operatorname{Sym}^2_{\mathbb{F}_q}(A) = \left(A \otimes_{\mathbb{F}_q} A\right) / \operatorname{span}\{\xi \otimes \eta - \eta \otimes \xi\}.$$

Consequently, the direct sum over the m_1 base forms has rank at most $m_1\binom{d+1}{2}$. After any public linear map to M outputs—in particular, for every relation-dependent matrix Q_R—the rank is at most

$$\min\left\{M, m_1\binom{d+1}{2}\right\}. \tag{10}$$

If the minimal polynomial of S has degree ℓ, then $d = \ell$ and, in the power basis,

$$z_{\mathrm{ord}}(u) = Dz_{\mathrm{sym}}(u). \tag{11}$$

Proof. Every element of A is symmetric. Since the displayed expression is a scalar,

$$u^\top \Sigma_\xi P_i \Sigma_\eta u = \left(u^\top \Sigma_\xi P_i \Sigma_\eta u\right)^\top = u^\top \Sigma_\eta P_i \Sigma_\xi u.$$

Thus every alternating label $\xi \otimes \eta - \eta \otimes \xi$ lies in the kernel. Taking the direct sum over i gives the first rank bound, and a linear map to M public outputs gives the additional codomain cap. When the powers of S form a basis of A, the factorization is exactly the coordinate identity (11). $\square$

Why the outer map cannot repair the loss. The public outer map receives only the values of these quadratic features. If two labeled inputs already define the same polynomial, no later linear combination can separate them. The outer map may have full rank on the K-dimensional quotient, but it cannot enlarge that quotient.

For $\ell = 4$, each base form loses six of its sixteen ordered coordinates: $\ell^2 = 16$ becomes $\binom{\ell+1}{2} = 10$. For $\ell = 2$, four ordered coordinates become three. These reductions occur for every relation and every symmetric public key.

The theorem is an upper bound: additional public dependencies could make the quadratic image even smaller. The concrete attacks separately check $\operatorname{rank} Q_R = K$ before solving. Thus the value K used in the cost tables is not inferred from symmetry alone; it is the exact rank observed in the public preflight for the chosen relation.

The factorization itself uses only transposition symmetry and is valid in every characteristic. Odd characteristic is needed only for the familiar direct-sum interpretation $A \otimes A = \mathrm{Sym}^2(A) \oplus \bigwedge^2(A)$. The submitted $q = 16$ system is outside this theorem because its public base matrices are not symmetrized.

5 Structured Stable Lifts

Proposition 1 permits arbitrary offsets v_j. Generic offsets are useful for making the affine equations full rank, but they create linear terms that mix all coordinates. That mixing destroys the block grading used by the selected published $\ell = 4$ special-XL model. We therefore need offsets that do two jobs at once:

(i) span the missing affine output directions, so that the full verifier is retained; and

(ii) confine all offset-linear terms to a small public row space, so that a large invariant kernel remains available for special XL.

The following construction achieves both. Choose one vector $w \in \mathbb{F}_q^{N_0}$ and multipliers $\Gamma_0, \dots, \Gamma_{r-1} \in A$, and set

$$v_j = \Sigma_{\Gamma_j} w. \tag{12}$$

All offsets are therefore generated from the same w by the public algebra A. Define the row space

$$\mathcal{U}_w = \operatorname{span}_{\mathbb{F}_q}\left\{w^\top \Sigma_\xi P_i \Sigma_\eta : 1 \le i \le m_1,\ \xi, \eta \in A\right\} \subseteq (\mathbb{F}_q^{N_0})^*. \tag{13}$$

This space contains every linear form that can arise from crossing a variable term with one of the structured offsets. Its a priori bound is $m_1 d^2$, not $m_1\binom{d+1}{2}$: one argument is the fixed offset vector w rather than the variable u on both sides, so reversing the two algebra labels need not leave this *linear* form unchanged.

A subspace $H \subseteq \mathbb{F}_q^{N_0}$ is called A-linear if $\Sigma_\xi H \subseteq H$ for every $\xi \in A$. Since $A \cong \mathbb{F}_{q^\ell}$, this is the analogue of extension-field linearity. To force such a kernel, take the complete A-orbit of the affine constraints. For a basis $1, S, \ldots, S^{d-1}$ of A, form

$$\mathcal{O}_{R,\mathbf{v}} = \begin{bmatrix} C_{R,\mathbf{v}} \\ C_{R,\mathbf{v}}\Sigma_S \\ \vdots \\ C_{R,\mathbf{v}}\Sigma_{S^{d-1}} \end{bmatrix}. \tag{14}$$

Theorem 3 (Stable-lift kernel)**.** *Retain the assumptions $S^\mathsf{T} = S$ and $P_i^\mathsf{T} = P_i$ from Theorem 2. For offsets of the form (12):*

(i) $\mathcal{U}_w$ is closed under right multiplication by A and $\dim_{\mathbb{F}_q} \mathcal{U}_w \le m_1 d^2$;

(ii) every row of $\mathcal{F}_{R,\mathbf{v}}$ and $\mathcal{O}_{R,\mathbf{v}}$ lies in $\mathcal{U}_w$;

(iii) if $\operatorname{rank} \mathcal{O}_{R,\mathbf{v}} = m_1 d^2$, then

$$H_{R,\mathbf{v}} = \ker \mathcal{O}_{R,\mathbf{v}}$$

is A-linear, has base-field dimension $N_0 - m_1 d^2$, and satisfies $\mathcal{F}_{R,\mathbf{v}} H_{R,\mathbf{v}} = 0$.

For the irreducible degree-ℓ construction, $H_{R,\mathbf{v}}$ has A-dimension $n - m_1\ell$.

Proof. See Appendix A.1. □

Interpretation. When the public orbit matrix has the stated rank, its kernel is a large search direction space closed under A. Moving inside this kernel changes the quadratic features but changes none of the offset-linear features. The offsets can therefore solve the missing affine equations, while the nonlinear solver sees a clean A-linear slice.

The stable-lift logic in four arrows. Structured offsets force every offset-linear row into $\mathcal{U}_w$; full orbit rank makes $\operatorname{row}(\mathcal{O}_{R,\mathbf{v}}) = \mathcal{U}_w$; therefore $H_{R,\mathbf{v}} = \ker \mathcal{O}_{R,\mathbf{v}}$ is annihilated by $\mathcal{F}_{R,\mathbf{v}}$; and the complete A-orbit in $\mathcal{O}_{R,\mathbf{v}}$ makes $H_{R,\mathbf{v}}$ A-linear. The offsets can thus cover the missing affine output directions without destroying the block structure used by special XL.

5.1 From an A-linear kernel to four variable blocks

Special XL needs variables divided into blocks with controlled multidegrees. The algebra A supplies those blocks after extending scalars to a field Ω in which the minimal polynomial of S splits. Over Ω, multiplication by S diagonalizes into ℓ eigenspaces. The primitive idempotent e_a is simply the projection onto the ath eigenspace. This scalar extension is a solver coordinate change, not a relaxation of the signature problem: the recovered point must still descend to $\mathbb{F}_q$ and pass the original verifier.

Let $\mathbf{e}_a \in \mathbb{Z}^\ell$ denote the ath standard unit vector. A quadratic using only block a has multidegree $2\mathbf{e}_a$; a bilinear term using blocks a and b has multidegree $\mathbf{e}_a + \mathbf{e}_b$.

Lemma 4 (Eigenblock coordinates of the quotient)**.** *Let $\Omega/\mathbb{F}_q$ be a splitting field of the minimal polynomial of S, and let $e_1, \ldots, e_\ell$ be the primitive idempotents of $A \otimes_{\mathbb{F}_q} \Omega$. If L is an A-linear space of A-dimension s, then*

$$L \otimes_{\mathbb{F}_q} \Omega = L_1 \oplus \cdots \oplus L_\ell, \qquad L_a = \Sigma_{e_a}(L \otimes_{\mathbb{F}_q} \Omega), \qquad \dim_\Omega L_a = s.$$

For each i, the eigenblock features

$$x^{\mathsf{T}}\Sigma_{e_a}P_i\Sigma_{e_b}x, \qquad 1 \le a \le b \le \ell,$$

are an invertible linear recombination of the symmetric power-pair features. They have multidegree $2\mathbf{e}_a$ when $a = b$ and $\mathbf{e}_a + \mathbf{e}_b$ when $a < b$.

Proof. Finite fields are perfect, so the irreducible minimal polynomial of S is separable. Over Ω, the power basis and the primitive-idempotent basis of $A \otimes \Omega \cong \Omega^\ell$ are related by an invertible Vandermonde matrix. Passing to the symmetric square gives an invertible change between the unordered power-pair features and the unordered eigenblock features. The idempotent e_a projects $L \otimes \Omega$ onto L_a. Hence a diagonal feature uses only variables in L_a, while an off-diagonal feature is bilinear in the variables from L_a and L_b, giving the stated multidegrees. □

Corollary 5 (Stable affine slices)**.** *Assume the public checks*

$$\operatorname{rank} Q_R = K, \qquad \operatorname{rank} C_{R,\mathbf{v}} = M - K, \qquad \operatorname{rank} \mathcal{O}_{R,\mathbf{v}} = m_1\ell^2 \tag{15}$$

pass, and let $L \subseteq H_{R,\mathbf{v}}$ be an A-linear subspace of A-dimension s (equivalently, base-field dimension $h = \ell s$). For every target, the affine equations in Proposition 1 have a solution space $a(t) + \ker C_{R,\mathbf{v}}$ for a publicly computed base point $a(t)$. For every $y \in \ker C_{R,\mathbf{v}}$, the affine slice $a(t) + y + L$ is contained in that space, and the full offset-linear feature $\mathcal{F}_{R,\mathbf{v}}u$ is constant on the slice.

After extension to a splitting field, use the eigenblock coordinates of Lemma 4. Translation by $a(t) + y$ creates only linear and constant terms in the same one or two blocks as the corresponding quadratic term. Introducing one homogenizing variable per block—an auxiliary variable that is set to one after solving—therefore turns each complete affine equation into a homogeneous equation of multidegree $2\mathbf{e}_a$ or $\mathbf{e}_a + \mathbf{e}_b$. If the restricted quadratic feature map has total rank K, then the disjoint monomial supports of these unordered multidegrees force rank m_1 in every component: there are $\binom{\ell+1}{2}$ components, each has rank at most m_1, and their ranks sum to $K = m_1\binom{\ell+1}{2}$.

Proof. See Appendix A.2. □

What the preflight certifies. The first two ranks in (15) say that the quotient has its full expected rank and that the offsets supply every remaining affine output direction. The orbit rank says that the kernel has the predicted A-linear dimension and kills all offset-linear motion. The restricted quadratic rank then certifies that no multidegree component was lost on the chosen slice. All of these tests are public and occur before the expensive Macaulay computation. A failing relation, offset family, or slice is simply replaced.

5.2 Passing the verifier's block-symmetry check

The public verifier does more than evaluate the quadratic map. It also rejects signatures containing too many symmetric blocks. A root of the reduced MQ system is useful only if it passes this final check. We therefore build rejection compatibility into the reduction rather than treating it as an unmodeled success probability.

Version 2.3 rejects a square odd-characteristic signature if one of its blocks is symmetric when $\ell > 2$, and rejects an $\ell = 2$ signature if more than $\lfloor n/4 \rfloor$ blocks are symmetric [18]. Rectangular blocks are not subject to this rule.

The structured $\ell = 4$ rows. For each square block b, substituting (5) into the equalities $U_b[a, c] = U_b[c, a]$ gives an affine system

$$G_b u = -h_b. \tag{16}$$

If $\operatorname{rank}[G_b \mid -h_b] > \operatorname{rank} G_b$, this system is inconsistent, so no point in the entire residual search space can make block b symmetric. Requiring this inconsistency for every square block is an exact public certificate that every root produced by the structured lift passes rejection.

A deterministic $\ell = 2$ construction. The first two parameter levels use offsets that force a fixed nonzero skew in every block.

Proposition 6 (Fixed-skew $\ell = 2$ lift)**.** *For the Version 2.3 $\ell = 2$ matrix*

$$S = \begin{pmatrix} 1 & 2 \\ 2 & 15 \end{pmatrix}, \qquad R_0 = I, \qquad R_1 = 9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix},$$

choose offsets satisfying

$$(v_1)_{b,0} - (v_0)_{b,1} = 1 \tag{17}$$

for every block b. Then every block of every signature satisfying the column relation has

$$U_b[0, 1] - U_b[1, 0] = 1. \tag{18}$$

Consequently every solution passes the $\ell = 2$ symmetric-block rejection check. If the public preflight also gives $\operatorname{rank} Q_R = K$ *and* $\operatorname{rank} C_{R,\mathbf{v}} = M - K$*, every target passes the affine consistency test and leaves K equations of degree at most two in $N_0 - M + K$ variables. Every root of that residual system is a valid signature.*

Proof. The first row of R_1 is the second row of $R_0 = I$. Hence the variable part of $U_b[0, 1] - U_b[1, 0]$ vanishes. The remaining offset difference is one by (17), proving (18). No block is symmetric. The final statement is the full-lift case of Proposition 1. □

Thus the fixed-skew construction has no rejection probability: every MQ root is acceptable. The Level-V headline attack uses a faster alternative in which targets are filtered before solving.

The zero-offset $\ell = 2$ filter. Set $R_0 = I$, $R_1 = 0$, and use zero offsets. Each block then has the form

$$\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix},$$

so it is symmetric exactly when $y = 0$. These rejection events are nonzero hyperplanes on disjoint block coordinates. This relation is chosen carefully: a relation that forced the two off-diagonal entries to agree would make every block symmetric and would be useless.

Let N be the residual-domain dimension ($N = N_0$ for this zero-offset lift) and let $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$ be the residual map of degree at most two. For each nonzero $b \in \mathbb{F}_q^K$, take the polar rank of the scalar quadratic $b \cdot g$, and let r_* be the minimum of these ranks. Thus r_* measures the worst output direction; a large r_* forces every target fiber to have nearly its expected size.

Put

$$t_{\text{rej}} = \lfloor n/4 \rfloor + 1, \qquad B = \binom{n}{t_{\text{rej}}}, \qquad \alpha = 1 - q^{-K},$$

and let $\mathcal{X}_{\text{acc}}$ be the assignments accepted by the block-symmetry filter. At least t_{rej} symmetric blocks are needed for rejection. Because the n hyperplanes use disjoint coordinates, the exact density of accepted assignments is

$$a_n := \frac{|\mathcal{X}_{\text{acc}}|}{q^N} = q^{-n} \sum_{k=0}^{\lfloor n/4 \rfloor} \binom{n}{k} (q-1)^{n-k}. \tag{19}$$

Proposition 7 (Accepted targets for the $\ell = 2$ filter)**.** *Every target has an accepted root if*

$$1 - Bq^{-t_{\text{rej}}} > \alpha(1+B) q^{K - r_*/2}. \tag{20}$$

For a uniform target $z \in \mathbb{F}_q^K$,

$$\Pr[g^{-1}(z) \cap \mathcal{X}_{\text{acc}} \neq \varnothing] \geq \frac{a_n}{1 + \alpha q^{K - r_*/2}}. \tag{21}$$

Moreover, if $1 - \alpha q^{K - r_*/2} > 0$, every target z satisfies

$$\frac{|g^{-1}(z) \cap \mathcal{X}_{\text{acc}}|}{|g^{-1}(z)|} \geq 1 - \frac{Bq^{-t_{\text{rej}}} + \alpha B q^{K - r_*/2}}{1 - \alpha q^{K - r_*/2}}, \tag{22}$$

whenever the right-hand side is positive.

Proof. See Appendix A.3. □

The three conclusions answer different questions. Readers interested only in the concrete attack may retain the following takeaway: for the Level-V system, the global condition $r_* \geq 240$ implies the first inequality and therefore gives an accepted root for every target. More generally, (20) is a sufficient worst-case condition: every target has an accepted root. Inequality (21) lower-bounds the fraction of uniformly sampled targets that have one. Inequality (22) says that, target by target, almost every root is accepted when the rank is high enough. The Level-V headline uses the first condition; the second gives an explicit retry factor if only a weaker global rank bound is available.

6 Root Existence, Uniqueness, and Retry Bounds

After the exact affine reduction, the structured attack repeatedly chooses a hash target and an affine coset of the selected slice direction space. We call one such pair a *target–slice trial.* The attack asks whether the resulting MQ system has a recoverable root. Three events must not be conflated:

(i) the slice contains at least one base-field root;

(ii) it contains exactly one base-field root; and

(iii) the chosen Macaulay matrix has a one-dimensional kernel from which that root can be read.

This section proves statements about the first two events. The third is the separate special-XL solving assumption in Section 7.2.

Let V be the full residual direction space, let $L \subseteq V$ be the chosen slice direction space, and write $h = \dim L$. For a map of degree at most two $g : V \to \mathbb{F}_q^K$, a random-map heuristic predicts

$$\mu = q^{h-K}$$

roots in a random affine coset of L above a random target. When $h = K - \tau$ with $\tau \geq 0$, this mean is $q^{-\tau}$, suggesting about q^{τ} trials. The obstruction detected by a second-moment analysis is excess collisions: distinct points in the same slice mapping to the same target. Polar derivatives count those collisions exactly, so the argument below is a theorem rather than a random-system heuristic. The cleanest case is full cross-polar rank: then distinct points behave almost as independently as the mean-root calculation suggests. For the positive slice deficits used here, a unique-root trial costs essentially $q^{K-h} = q^{\tau}$ attempts. The theorem first states the exact formulas and then quantifies what changes when some output directions lose rank.

Write the vector-valued polar map as

$$\beta(x, y) = g(x + y) - g(x) - g(y) + g(0).$$

For $b \in \mathbb{F}_q^K \setminus \{0\}$, define the cross-polar map

$$\Phi_{b,L} : V \longrightarrow L^*, \qquad \Phi_{b,L}(x)(y) = b \cdot \beta(x, y), \tag{23}$$

and let $e_L(b) = \operatorname{rank} \Phi_{b,L}$. Put

$$e_* = \min_{b \neq 0} e_L(b), \qquad \Theta_L(g) = \sum_{b \neq 0} q^{-e_L(b)}. \tag{24}$$

For $y \in L \setminus \{0\}$, also define the directional derivative

$$T_y : V \longrightarrow \mathbb{F}_q^K, \qquad T_y(x) = \beta(x, y), \tag{25}$$

write $r(y) = \operatorname{rank} T_y$, and let $\kappa(y)$ be one when $-(g(y) - g(0)) \in \operatorname{im} T_y$ and zero otherwise. The indicator $\kappa(y)$ records whether two points separated by y can collide at all; the rank $r(y)$ then determines how many such pairs exist.

Theorem 8 (Root collisions and the rank-spectrum bound)**.** *Choose an affine coset C uniformly from V/L and choose $z \in \mathbb{F}_q^K$ uniformly. Let*

$$Z = |\{x \in C : g(x) = z\}|, \qquad \mu = q^{h-K},$$

and define

$$\eta_0 = \sum_{y \in L \setminus \{0\}} \kappa(y) q^{-r(y)}, \qquad \bar{\eta} = \sum_{y \in L \setminus \{0\}} q^{-r(y)}. \tag{26}$$

Then

$$\mathbb{E}Z = \mu, \tag{27}$$

$$\mathbb{E}[Z(Z-1)] = \mu\eta_0, \tag{28}$$

$$\eta_0 \leq \bar{\eta} = \mu(1 + \Theta_L(g)) - 1. \tag{29}$$

Consequently,

$$\operatorname{Var}(Z) \leq \mu^2 \Theta_L(g), \tag{30}$$

$$\Pr[Z > 0] \geq \frac{\mu}{1 + \eta_0} \geq \frac{1}{1 + \Theta_L(g)}. \tag{31}$$

If $h = K - \tau$ and $e_* \geq h - \Delta$, then

$$\Theta_L(g) < q^{\tau+\Delta}, \qquad \mathbb{E}[\text{trials to a root-containing target–coset pair}] < 1 + q^{\tau+\Delta}. \tag{32}$$

This last bound controls the search for a root-containing target–coset pair, not uniqueness of the root. Here Δ is the maximum loss from full cross-polar rank h.

Proof. See Appendix A.4. ∎

How to read the theorem. The first moment $\mathbb{E}Z = \mu$ is exactly the random-map value. The second factorial moment $\mathbb{E}[Z(Z-1)]$ counts ordered pairs of distinct roots and therefore measures collisions. The compatible collision mass η_0 is exact; $\bar{\eta}$ drops the compatibility test and is a convenient upper bound. The sum $\Theta_L(g)$ describes the entire cross-polar rank spectrum, not merely its minimum. A few isolated rank defects may have little effect, whereas low rank in many output directions can substantially increase retries.

$\mu = q^{h-K}$	expected number of roots in a random target–slice pair
η_0	exact expected number of compatible collisions per planted root
$\bar{\eta}$	rank-only upper bound on that collision mass
full cross-polar rank	for the selected positive deficits, unique-root trial count essentially $q^{K-h} = q^{\tau}$

Corollary 9 (Unique-root trials and the planted distribution)**.** *In the setting of Theorem 8,*

$$\Pr[Z = 1] \geq \mu(1 - \eta_0) \geq \mu(1 - \bar{\eta}), \tag{33}$$

$$\Pr[Z > 0] \geq \mu\left(1 - \frac{\bar{\eta}}{2}\right), \qquad (\bar{\eta} < 2), \tag{34}$$

$$\Pr[Z \geq 2 \mid Z > 0] \leq \frac{\bar{\eta}}{2 - \bar{\eta}}, \qquad (\bar{\eta} < 2). \tag{35}$$

In particular, if $\bar{\eta} < 1$, the expected number of independent trials until $Z = 1$ is at most

$$\frac{1}{\mu(1 - \bar{\eta})}. \tag{36}$$

Let U be the uniform target–coset pair conditioned on $Z > 0$, and let P be the planted distribution obtained by choosing a uniform coset, a uniform point in it, and the point's image as target. If $\bar{\eta} < 1$, then

$$\Pr_{\mathsf{P}}[Z \geq 2] \leq \eta_0 \leq \bar{\eta}, \qquad d_{\mathrm{TV}}(\mathsf{U}, \mathsf{P}) \leq \bar{\eta}. \tag{37}$$

Proof. See Appendix A.5. $\square$

Total variation distance is the largest discrepancy that the two distributions can assign to any event. Thus, when $\bar{\eta}$ is tiny, an experiment that plants a known root samples essentially the same target–coset distribution as a uniformly chosen trial conditioned on being solvable. This justifies using planted instances to probe solver behavior; it does not remove the separate finite-size-to-production extrapolation.

Corollary 10 (Full cross-polar rank)**.** *The following are equivalent:*

(i) $e_L(b) = h$ *for every* $b \neq 0$*;*

(ii) T_y *is surjective for every* $y \in L \setminus \{0\}$*.*

Under these conditions,

$$\eta_0 = \bar{\eta} = \mu - q^{-K}.$$

If $h = K - \tau$ with $\tau \geq 0$, the expected number of trials until a unique base-field root is at most

$$\mathsf{Retry}_{K,\tau} := \frac{q^{\tau}}{1 - q^{-\tau} + q^{-K}}. \tag{38}$$

Proof. If some T_y is not surjective, a nonzero b annihilates its image, so the transpose of $\Phi_{b,L}$ has the nonzero kernel vector y and $e_L(b) < h$. The converse is the same argument in reverse. Under either equivalent condition, $r(y) = K$ and $\kappa(y) = 1$ for every nonzero y, giving $\eta_0 = \bar{\eta} = (q^h - 1)q^{-K} = \mu - q^{-K}$. Substitute this value into (36). □

For the positive deficits used in the concrete attacks, the expected unique-root trial count under full cross-polar rank is essentially q^τ: the denominator in (38) differs from one only by the tiny terms $q^{-\tau}$ and q^{-K}. In the running Level-I example, $h = 40$, $K = 50$, and $\tau = 10$, so the expected number of independent target–slice trials is essentially $19^{10} = 2^{42.48}$. Conditional on a root-containing trial, the probability of more than one base-field root is below $2^{-43.47}$. This is the precise justification for the retry factor used in the structured cost model.

Equation (29) distinguishes sparse rank defects from a pervasive loss. If at most T nonzero coefficient vectors b are defective and each has defect $h - e_L(b) \le \Delta$, then

$$\bar{\eta} \le \mu - q^{-K} + Tq^{-K}(q^\Delta - 1). \tag{39}$$

If defects are counted projectively, T in this formula must include all $q - 1$ nonzero scalar multiples of each projective direction. Under full cross-polar rank and $q = 19$, (35) is below $2^{-43.47}$ for $\tau = 10$ and below $2^{-60.47}$ for $\tau = 14$; the corresponding total-variation bounds are below $2^{-42.47}$ and $2^{-59.47}$.

Corollary 11 (Full-space target availability)**.** *Let $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$ have degree at most two. Let r_* be the minimum polar rank of a nonzero output combination, and put $\alpha = 1 - q^{-K}$. For a uniform target,*

$$\Pr[g^{-1}(z) \neq \varnothing] \ge \frac{1}{1 + \alpha q^{K - r_*}}. \tag{40}$$

Hence the expected number of target trials is at most $1 + \alpha q^{K-r_*}$. If $r_* \ge 2K$, every target has a root.

Proof. See Appendix A.6. □

For a structured $\ell = 4$ lift, an invertible output transformation separates the $M - K$ affine coordinates from the remaining K coordinates of a uniform target. Conditioning on the affine coordinates fixes a base point; the other coordinates remain uniform, and translation does not change the polar maps. We take $V = \ker C_{R,\mathbf{v}}$ and let L be the selected stable slice, of dimension $h = \ell s = K - \tau$. Full cross-polar rank gives the unique-root retry bound $\mathsf{Retry}_{K,\tau}$ in (38). A worst-case rank-defect bound gives the budget for reaching a root-containing target–coset pair in (32); unique recovery instead depends on the aggregate quantity $\bar{\eta}$ and the Macaulay kernel model.

For the fixed-skew $\ell = 2$ lift, take $L = V$ to be the complete residual space. By Proposition 6, every root is accepted. The expected target-search factor is at most $1 + \alpha q^{K-r_*}$, and $r_* \ge 2K$ makes every target solvable. Under the AXN conversion, the arithmetic budgets remain below the nominal NIST references for global ranks at least 46, 68, and 89 at Levels I, III, and V. The sufficient all-target thresholds are 96, 144, and 192. These are global conditions and concern the converted operation model, not a complete solver circuit; the sampled evidence in Section 8 is not a certificate. The Level-V filter attack instead uses Proposition 7.

6.1 Sampling exactly uniform targets with the official conversion

The probability theorems above assume a uniform target in $\mathbb{F}_q^K$. Reducing binary hash chunks modulo 19 introduces a small bias, so uniformity cannot simply be assumed. The attacker removes it without changing the verifier: it repeatedly chooses a message–salt input, runs the official hash-to-field conversion, and keeps the input only when each underlying binary chunk lies in a range that makes the resulting base-19 digits exactly uniform.

Concretely, the official $q = 19$ conversion reads at most fifteen base-19 digits from each 64-bit chunk. If a uniform b-bit chunk X is to supply a digits, the attacker accepts that message–salt candidate only if

$$X < \left\lfloor \frac{2^b}{q^a} \right\rfloor q^a. \tag{41}$$

Conditioned on this event, $X \bmod q^a$ is uniform, so the extracted digits are jointly uniform. Applying (41) independently to every chunk makes the retained official target exactly uniform. The invertible output-coordinate change used in Section 3.2 preserves uniformity. Conditioning on the values of the $M - K$ affine-output coordinates—or on their passing values in a filtered lift—then leaves the remaining K residual target coordinates exactly uniform, which is the distribution assumed in the root theorems. This is ordinary rejection sampling performed by the forger over message–salt candidates; the verifier continues to apply the unmodified official conversion. Across all Version 2.3 lengths the acceptance probability is at least 0.152462, or fewer than 6.6 candidates on average.

The Level-V $\ell = 2$ filter needs about $19^{32} > 2^{128}$ raw targets before the $M - K = 32$ consistency conditions pass, so the 128-bit salt alone does not supply enough distinct inputs. An existential forger may vary a chosen-message prefix as well as the salt. Distinct prefix–salt pairs are distinct random-oracle inputs and therefore give independent targets in the model. The artifact's deliberately loose target-generation envelope is below 2^{32} gates per target retained by the unbiased hash-chunk sampler (the AXN charge is no larger than the NAND envelope used there). Even after multiplication by the 19^{32} filtering factor, this term is more than 70 exponent bits below the cost-driving MQ solve.

7 Concrete Costs

The algebraic reduction determines the residual MQ systems exactly. Turning those systems into numerical security estimates requires a separate cost model. We report that model in layers so that a reader can see which numbers are structural, which come from an MQ solver estimate, and which are merely a change of arithmetic units.

Table 4: How to interpret the cost accounting.

Layer	What it means
Residual system	Exact numbers of equations, variables, blocks, and public rank conditions after the reduction.
Solver model	Published generic-MQ or special-XL formula for the dominant finite-field work at those dimensions.
AXN conversion	Replacement of the solver formula's coarse field-operation factor by an explicit two-input AND/XOR/XNOR circuit for one signed multiply-accumulate.
Not included	A complete implementation of sparse matrix construction, pivoting and control, memory addressing and traffic, communication, or an optimized end-to-end attack netlist.

A cost exponent c means modeled work proportional to 2^c in the stated unit. A difference of ten exponent bits is a factor of about 2^{10}, not ten percent. Because the solver formulas count dominant arithmetic rather than a complete machine, we call the difference from a NIST reference an *arithmetic overhead budget*.

7.1 Cost units and circuit conversion

The cited attacks first count finite-field operations and then apply coarse bit-cost factors [2, 4]. NIST lists 2^{143}, 2^{207}, and 2^{272} as nominal classical-gate references, while also requiring category comparisons across every metric it considers relevant to practical security [15]. We retain each attack's field-operation count and replace only its dominant field factor by explicit circuits. The resulting figures sharpen one metric; they do not by themselves establish a category break. Our primary AXN basis assigns unit cost to two-input AND, XOR, and XNOR gates, with free constants, wires, fan-out, and structural sharing. This is the gate family used by the concrete AES-128 circuit on NIST's circuit list [13]. A two-input-NAND conversion provides a separate basis-sensitivity check.

For generic MQ, the published factor is

$$b_1 = (\log_2 19)^2 + \log_2 19 = 22.29. \tag{42}$$

Using canonical five-bit representatives and exact Barrett reduction, the validated circuits for one signed multiply-accumulate, $c + ab$ or $c - ab$, over $\mathbb{F}_{19}$ use 402 and 441 AXN gates, respectively; their NAND translations use 781 and 835 gates. To avoid assuming a particular sign schedule in elimination, we charge

$$g_1^{\mathrm{AXN}} = 441, \qquad g_1^{\mathrm{NAND}} = 835. \tag{43}$$

Both signs are checked exhaustively on all 19^3 valid input triples by the generator and by an independent scalar evaluator.

For special XL, we work in the isomorphic sparse basis $\mathbb{F}_{19}[t]/(t^4 - t - 1)$. In this basis, the element $13 + 13t + 15t^2 + 2t^3$ is a root of the official defining polynomial; evaluating its powers supplies the explicit basis change to $\mathbb{F}_{19}[S]$. Two-level Karatsuba uses nine base-field multiplications, and $t^4 = t + 1$ makes reduction additive. The validated circuits for $c \pm ab$ over $\mathbb{F}_{19^4}$ use 6,077 and 6,081 AXN gates, or 11,086 and 11,102 NAND gates. We therefore charge

$$g_4^{\mathrm{AXN}} = 6{,}081, \qquad g_4^{\mathrm{NAND}} = 11{,}102. \tag{44}$$

The artifact checks the complete scalar basis tensor, 4,096 random extension products, the basis isomorphism, and both translated netlists. Charging a signed multiply-accumulate for every multiplication in the special-XL formula is conservative.

Same-basis AES calibration. The release builds variable-key AES-128/192/256 one-pair key-filter circuits with the NIST 113-gate S-box, the complete key schedule, encryption, and a final ciphertext comparator. To identify the actual key rather than merely a key that matches one pair, surviving candidates are checked lazily on enough independent plaintext–ciphertext pairs to leave at most 2^{-128} expected wrong survivors in the ideal-cipher model: a total of two pairs for AES-128 and a total of three for AES-192 and AES-256. The additional average work is negligible because only survivors reach those checks. Every first-pair circuit is validated on the FIPS test vector and eight additional keys against an independent AES implementation.

The NIST-listed 28,600-gate AES-128 circuit has full-key-space exponent $128 + \log_2(28{,}600) = 142.80$, numerically consistent with the nominal value 143. Across our nine rows, the AXN arithmetic estimates are 9.13–53.86 bits below the constructive full-space AES circuits and 9.25–54.52 bits below the nominal NIST references. The NAND cross-check gives attack ranges $2^{133.04}$–$2^{134.67}$, $2^{175.91}$–$2^{188.67}$, and $2^{218.35}$–$2^{240.42}$.

Table 5: Constructive AES calibration circuits. Full charges the entire key space for the first-pair filter; expected search is one bit lower, and the lazy confirmation overhead is negligible under the stated ideal-cipher model. These circuits are not lower bounds on optimal AES implementations and do not replace NIST's nominal references.

Level	pairs	NIST nominal	AXN gates/filter	AXN full	NAND gates/filter	NAND full
I	2	143	30,081	142.88	106,616	144.70
III	3	207	34,227	207.06	121,735	208.89
V	3	272	41,543	271.34	147,772	273.17

Scope of the conversion. The AXN and NAND figures price the field-arithmetic schedule counted by the published estimators; they do not include solver control, memory traffic, or a complete sparse-linear-algebra implementation. Public preprocessing and exact-uniform target generation are bounded separately, so the dominant unresolved inputs are the global production rank spectra and the production-scale solving models.

7.2 Structured special XL for $\ell = 4$

The exact reduction supplies an A-linear slice of A-dimension s, hence base-field dimension $4s$ when $\ell = 4$. After extension to the splitting field, this becomes four blocks of s variables. For each unordered pair of blocks there are m_1 independent equations of degree two: diagonal equations use one block twice, and off-diagonal equations use two blocks once each. Special XL records a separate degree for each block and selects a multidegree at which the associated Macaulay matrix is predicted to have the needed rank. A reader need not manipulate the series below: here it is only a compact device for choosing the matrix degree and counting the resulting monomial columns.

After adding one homogenizing variable to each block, the Hilbert series used by the published special-XL model is

$$H_{m_1,s}(\mathbf{t}) = \frac{\prod_{a=1}^{4}(1-t_a^2)^{m_1}\prod_{1\le a<b\le 4}(1-t_a t_b)^{m_1}}{\prod_{a=1}^{4}(1-t_a)^{s+1}}. \tag{45}$$

For $\mathbf{d} = (d_1, \ldots, d_4)$, put

$$M_s(\mathbf{d}) = \prod_{a=1}^{4}\binom{s+d_a}{d_a}. \tag{46}$$

The search chooses a multidegree whose signed Hilbert coefficient is negative. This sign change is a rank-prediction trigger in the cited special-XL model: a random Macaulay matrix is predicted to have full column rank, while a system with a planted solution is predicted to have a one-dimensional right kernel [4]. The modeled linear-algebra work for one target–slice trial is $3s^2\big(M_s(\mathbf{d})\big)^2$ extension-field multiplications: the square is the dense linear-algebra term in the published model, and $3s^2$ is its stated prefactor.

Cabarcas et al. handle an underdetermined system by specializing k coordinates, where $4 \mid k$, and multiplying the work by q^k. Our attack uses a different retry loop: it fixes a $4s$-dimensional slice and samples the target and affine coset. Under full cross-polar rank, Corollary 10 bounds the expected trials needed for a unique base-field root by $\mathsf{Retry}_{K,\tau}$. The modeled work is therefore

$$\mathsf{Retry}_{K,\tau}\, 3s^2 M_s(\mathbf{d})^2, \qquad \tau = K - 4s, \tag{47}$$

extension-field multiplications. Replacing the older $1 + q^\tau$ factor by $\mathsf{Retry}_{K,\tau}$ changes each displayed logarithm by less than 10^{-25} bits. If every cross-polar map has rank defect at most Δ, then (32) bounds the expected target–coset trials—and hence the associated Macaulay linear-algebra budget under the same per-trial model—by $1 + q^{\tau+\Delta}$. It does not by itself imply unique recovery. That step depends on $\bar{\eta}$ and on the Macaulay kernel behavior described below.

The computation takes place over $\mathbb{F}_{19^4}$. The published convention assigns each extension-field multiplication

$$b_4 = 2\big(\log_2 19^4\big)^2 + \log_2 19^4 = 594.43 \tag{48}$$

bit operations. The AXN arithmetic conversion replaces b_4 by the conservative signed multiply-accumulate bound g_4^{AXN} from (44).

Lemma 12 (Recovery from a one-dimensional Macaulay kernel)**.** *Fix a target t, a selected affine slice $a(t) + y + L$, and an $\mathbb{F}_q$-linear parameterization $\iota : \mathbb{F}_q^h \to L$. Let $\lambda_* \in \mathbb{F}_q^h$ be slice coordinates for a root*

$$u_* = a(t) + y + \iota(\lambda_*),$$

and let $x_* \in \Omega^h$ be the image of λ_* under the eigenblock change of variables. Let $\mathcal{M}$ be the corresponding special-XL Macaulay matrix over the splitting field Ω, dehomogenized by setting every block homogenizer to one. Suppose its columns include the constant and degree-one monomials. If $\ker_\Omega \mathcal{M}$ is one-dimensional, row reduction recovers x_*, λ_*, the common column u_*, and hence the complete signature.

Proof. The dehomogenized monomial-evaluation vector $\nu(x_*)$ lies in $\ker_\Omega \mathcal{M}$ and has constant coordinate one. Every nonzero kernel vector is therefore a scalar multiple of $\nu(x_*)$. Normalizing its constant coordinate to one recovers x_* from the degree-one coordinates. The inverse eigenblock transformation recovers $\lambda_* \in \mathbb{F}_q^h$; the affine embedding gives $u_* = a(t) + y + \iota(\lambda_*)$; and the column relation (5) reconstructs every signature column. $\square$

Las Vegas wrapper. For each target–coset trial, compute the Macaulay kernel and continue unless it is one-dimensional with nonzero constant coordinate. Normalize that coordinate, recover the degree-one eigenblock coordinates, and invert the eigenblock transformation. Continue again unless the point lies in $\mathbb{F}_q$ and the public verifier accepts it. The procedure therefore never returns an invalid signature; the solving model predicts only how often a trial reaches the recoverable case with a one-dimensional right kernel.

The target–coset theorem counts $\mathbb{F}_q$-rational roots, so a unique-root trial contains the point required by the lemma. Corollary 9 bounds multiple-root trials and the distance from the planted-root distribution using the full cross-polar rank spectrum. Under full cross-polar rank, the conditional multiple-root probability is below $2^{-43.47}$ for $\tau = 10$ and below $2^{-60.47}$ for $\tau = 14$.

The remaining assumption that the production Macaulay right kernel is one-dimensional concerns extra algebraic kernel vectors, including those caused by extension-field roots or by the Macaulay rows failing to span all expected relations; it is not an uncharged descent retry. Every recovered point is checked by the original verifier. The small-profile tests in Section 8 support, but do not prove, this production solving model.

Precisely what the displayed special-XL costs assume. The exponents in Table 6 use (i) global full cross-polar rank on the selected slice, so that $\mathsf{Retry}_{K,\tau}$ bounds unique-root trials, and (ii) the production Macaulay behavior just described: a planted-solvable trial has a one-dimensional right kernel. A global rank-defect bound alone controls only retries to a root-containing target–coset pair through (32); it does not imply unique recovery or a one-dimensional right kernel. This separation is important because the first condition concerns an entire family of nonzero output

Table 6: Cost-driving special-XL parameters for the six $\ell = 4$ rows. "Model" uses b_4; "AXN arithmetic" replaces it by the validated signed multiply-accumulate circuit g_4^{AXN}.

m_1	available $\dim_A H$	K	(s, τ)	$\mathbf{d}$	model	AXN arithmetic
5	12–13	50	$(10, 10)$	$(3, 3, 3, 4)$	128.82	132.17
7	15–19	70	$(15, 10)$	$(4, 4, 6, 6)$	171.69	175.04
9	22–23	90	$(19, 14)$	$(5, 6, 6, 6)$	214.12	217.48

directions (modulo scalar multiples), whereas the second concerns production-scale algebraic solving behavior.

The exact search checks every admissible s and all 42,356 sorted multidegrees whose modeled work could match or beat the incumbent, including highly unbalanced degrees. It recovers the three optima in Table 6, uniquely up to block permutation. The conservative target-generation charge does not change the displayed hundredths.

For the search for a root-containing target–coset pair and the associated Macaulay linear algebra, the AXN budget retains a positive nominal arithmetic budget for $\Delta \leq 2, 7, 12$ at Levels I, III, and V. These thresholds neither price a complete solver circuit nor certify unique recovery. The latter additionally requires $\bar{\eta} < 1$ in (26) and the modeled production Macaulay matrix to have a one-dimensional right kernel.

7.3 The $\ell = 2$ attacks

For Levels I and III, use the fixed-skew lift of Proposition 6. The public preflight gives quotient rank K, affine rank $M - K$, and residual systems of dimensions 48/112 and 72/168. CFXL is a hybrid strategy that specializes some variables and then applies XL-style linear algebra; Hashimoto refines this optimization for underdetermined MQ. Minimizing these variants in Beullens's printed Appendix-A convention gives the per-solve estimates

$$2^{129.44} \quad \text{and} \quad 2^{183.44} \tag{49}$$

modeled bit operations [2, 6]. Replacing b_1 by g_1^{AXN} gives arithmetic exponents 133.75 and 187.74. Every recovered root is accepted. The headline per-solve estimates incur no target-retry factor when the global all-target conditions $r_* \geq 96$ and $r_* \geq 144$ hold. For a smaller global rank, the expected-cost exponent increases by at most $\log_2(1 + (1 - 19^{-K})19^{K - r_*})$. The same fixed-skew construction at Level V gives a 96/224 fallback with model exponent 237.84 and AXN exponent 242.15; its all-target threshold is $r_* \geq 192$.

The optimizer includes Hashimoto's published boundary value $a = 1$. For $\mathrm{MQ}(19, 1, 1)$, the semi-regular Hilbert-series rule used by CFXL never produces the stopping degree at which its linearization would be costed. The artifact therefore solves that elementary system by 19 direct evaluations and deliberately overcharges each evaluation. This lowers six generic quotient baselines and the Level-V fixed-skew fallback; it does not change the selected Level-I or Level-III fixed-skew costs.

The Level-V headline instead uses $R_0 = I, R_1 = 0$ and zero offsets. The quotient has rank K, target filtering enforces the $M - K$ public consistency conditions, and the residual system has 96 equations in all 256 variables. Target search occurs before equation solving, so its cost adds to—rather than multiplying—the cost of a solve. The generic-MQ solve has AXN arithmetic exponent 239.50, while exact-uniform target filtering is smaller by more than 70 exponent bits. The displayed exponent incurs no target-retry factor under the accepted-root condition (20); for this row,

Table 7: $\ell = 2$ systems and global polar-rank thresholds. Model is the pinned published operation model; AXN uses $g_1^{\mathrm{AXN}} = 441$. "For budget" is the smallest rank for which expected AXN arithmetic remains below the nominal NIST reference. "All-target root" is sufficient to guarantee that every target has an accepted root; it does not assert that one particular solver invocation succeeds.

Parameters	mode	eqs./vars.	model	AXN	for budget	all-target root
$(48, 16, 2, 2)$	fixed-skew	48/112	129.44	133.75	46	96
$(72, 24, 2, 2)$	fixed-skew	72/168	183.44	187.74	68	144
$(96, 32, 2, 2)$	filter	96/256	235.19	239.50	177	240
$(96, 32, 2, 2)$	fixed-skew	96/224	237.84	242.15	89	192

$r_* \geq 240$ suffices. At lower rank, (21) gives the additional expected retry factor. The fixed-skew fallback is slower but makes every recovered root acceptable by construction.

For the filter row, the all-target-root threshold includes rejection through (20); for fixed skew, it is the simpler condition $r_* \geq 2K$ from Corollary 11. The quotient matrix depends only on the fixed public outer map and the relation, not on a generated key. Exhaustive enumeration of all $19^2 = 361$ relations $R_1 = aI + bS$ proves quotient rank K for every relation on all three parameter shapes. Exhaustive optimization of the natural two-block special-XL model gives bit-operation exponents 132.31, 184.27, and 236.36, all worse than the selected generic-MQ modes.

8 Implementation and Evidence

The artifact implements the public key generator, verifier-side feature maps, symmetry quotient, affine reduction, structured lift, rank tests, and cost searches over $\mathbb{F}_{19}$. It does *not* report a production-size end-to-end forgery, a production-size Macaulay solve, or a wall-clock attack timing. Its full-size experiments concern public preprocessing and rank; its solver experiments use smaller tractable profiles. The checks therefore have different logical force. The table below is the intended reading guide for every experimental statement in this section.

Table 8: Evidence hierarchy. "Exact" here means exact for the concrete object checked, not an exhaustive theorem about all production directions.

Check	What is computed	What it does—and does not—establish
Public preflight	Exact ranks and rejection conditions for the chosen key, relation, offsets, and slice	The exact reduction is valid for that concrete attack instance.
Rank campaign	Exact ranks for many selected nonzero output or direction vectors	Finite evidence about the unenumerated global minimum rank; not a certificate for all directions.
Macaulay experiment	Exact kernels at smaller tractable dimensions	Evidence for the production solver model; not a production timing or rank proof.
Circuit validation	Exhaustive or independent checking of the released field primitive	Exact cost and functionality of that primitive, but not of a complete solver.

A "projective direction" below means a nonzero coefficient vector modulo multiplication by a nonzero field scalar; all scalar multiples define the same polar rank. Sampling many distinct

Table 9: Exactly computed cross-polar ranks for the tested structured $\ell = 4$ slice–direction pairs.

Parameters	slices	$h = 4s$	rank tests	sampled minimum
$(28, 5, 4, 4)$	3	40	26,296	40
$(28, 4, 4, 5)$	2	40	24,198	40
$(40, 7, 4, 4)$	2	60	45,658	60
$(38, 5, 4, 5)$	2	60	2,188	60
$(50, 9, 4, 4)$	2	76	2,228	76
$(52, 6, 4, 6)$	2	76	2,228	76
Total	13	–	102,796	maximal

directions can expose low-rank structure, but only exhaustive enumeration or a proof can certify the global minimum required by the headline all-target-root conditions.

8.1 Public preflight and verifier regression

For a structured $\ell = 4$ instance, the preflight checks

$$\operatorname{rank} Q_R = K, \qquad \operatorname{rank} C_{R,\mathbf{v}} = M - K, \qquad \operatorname{rank} \mathcal{O}_{R,\mathbf{v}} = m_1 \ell^2,$$

together with $\mathcal{F}_{R,\mathbf{v}} H_{R,\mathbf{v}} = 0$, the predicted A-kernel dimension, A-invariance, the restricted quadratic rank, and rejection compatibility. All 121 regenerated full-size instances tested in the artifact pass, including the official Level-I known-answer key. The first deterministic offset candidate passes for 114 instances; the second passes for the remaining seven.

For fixed skew, the preflight checks the relation and rejection identity and the quotient, affine, and augmented ranks. All 60 full-size keys tested in the artifact pass. The artifact also reproduces the official Level-I public key byte for byte, verifies its known-answer signature against the official hash target, and checks the fixed output-order permutation described in Section 3.2. Eighteen additional regressions cover all nine Version 2.3 shapes. An independent Gauss–Jordan implementation agrees with the rank kernel on 2,036 matrices, including 36 production-sized matrices.

8.2 Rank evidence

Structured slices. Every selected slice is tested on all coefficient-basis directions (one nonzero output coefficient at a time) and 1,024 deterministic dense directions (many nonzero coefficients). Selected slices also receive complete normalized weight-two tests (two nonzero coefficients, modulo overall scaling) or additional dense tests. All 102,796 tested cross-polar maps have maximal rank $h = 4s$; the per-shape counts are in Table 9. The dual campaign evaluates 21,992 directional maps T_y on the same 13 slices, and every one is surjective of rank K. All 124,788 coefficient and direction vectors used in these two campaigns are projectively distinct within their respective slice records.

Fixed-skew systems. A broad campaign tests every coefficient-basis direction together with combinations having two, three, or many nonzero output coefficients. A second campaign targets the 21 exceptional projective directions that make the local operator on $\mathrm{Mat}_2(\mathbb{F}_{19})$ rank two rather than four. It tests all 381 local projective directions in one block, every exceptional direction in every remaining block, and deterministic two-block exceptional combinations. Together the campaigns perform 3,888 exact rank evaluations; their sampled minima and the rigorous thresholds used by the cost analysis appear in Table 10.

Table 10: Full-size polar-rank evidence for the fixed-skew $\ell = 2$ systems. "For budget" is the minimum global polar rank that preserves positive AXN arithmetic budget in Table 7; "all-target root" guarantees root existence for every target, not success of a particular solver run.

Parameters	preflights	rank tests	sampled min.	for budget	all-target root
$(48, 16, 2, 2)$	20	1,104	110	46	96
$(72, 24, 2, 2)$	20	1,296	167	68	144
$(96, 32, 2, 2)$	20	1,488	223	89	192
Total	60	3,888	–	–	–

Zero-offset filter. The broad and exceptional-direction campaigns perform 29,424 exact rank evaluations on two keys per shape. Their sampled minima are 127, 190, and 254 in dimensions 128, 192, and 256. For the Level-V filter, the sampled minimum 254 is above the sufficient accepted-root threshold 240. If the global minimum is at least 254, then (22) gives an accepted fraction of at least 0.999999999 in every target fiber. The experiment supports that hypothesis but does not certify the global minimum.

8.3 Solver and probability checks

The special-XL search examines all 42,356 admissible multidegrees at modeled work no greater than the incumbent and recovers the three optima in Table 6. At tractable sizes, the artifact builds 128 four-block Macaulay matrices: 64 random instances have full column rank and 64 planted-solvable instances have a one-dimensional right kernel, with the planted evaluation vector in that kernel. A 48-matrix two-block control contains one planted case with a two-dimensional right kernel, so the corresponding $\ell = 2$ special-XL estimates are not used. These are generic small-profile experiments, not production SNOVA timings.

Exhaustive checks over $\mathbb{F}_3$ and $\mathbb{F}_5$ verify the exact first and factorial second moments, the compatible-collision formula, the rank-spectrum upper bound, and the singleton and planted-distribution inequalities on 110 complete quadratic maps. Seventeen maps have nonmaximal cross-polar rank; in twelve, compatibility makes η_0 strictly smaller than $\bar{\eta}$. Finally, on six actual cost-driving $\ell = 2$ cores, the homogeneous quadratic outputs are linearly independent, the common polar radical—the set of directions annihilated by every polar form—is zero, and all 384 sampled homogeneous Jacobians (matrices of first derivatives of the homogeneous quadratic parts) attain the maximum rank permitted by their dimensions. These are genericity diagnostics, not solving-degree measurements.

8.4 Scope of the estimates

Once a public preflight passes, the quotient, affine reduction, stable kernel, fixed-skew identity, and rejection handling are exact. The remaining production-scale inputs are the global polar spectra and the solving behavior. The sampled ranks are broad but finite evidence and do not enumerate the full families of nonzero output directions (modulo scalar multiples). Likewise, the displayed exponents use the published special-XL and generic-MQ models rather than full-size Gröbner or XL runs; no production-size end-to-end solve is claimed. The artifact checks every dimension, Hilbert coefficient, public rank, and cost supplied to those models and verifies every recovered candidate with the original public verifier.

What would turn the estimates into end-to-end certificates. The rank assumptions would require either an algebraic proof of the global minimum over every nonzero output direction or a complete enumeration enabled by additional structure. The solver assumptions would require production-size Macaulay or Gröbner runs, or a validated implementation whose observed ranks and costs replace the semi-regular models. A complete gate comparison would additionally need to price control flow, pivoting, indexing, memory, and data movement.

9 Countermeasures

The design lesson is that security must be based on distinct quadratic polynomials, not on the number of ordered labels used to name them. A security analysis for any symmetric redesign should compute the actual quotient rank ρ_{quad} and use the unconditional cap

$$\rho_{\text{quad}} \le K_d = \min\left\{M, m_1\binom{d+1}{2}\right\}, \qquad d = \dim_{\mathbb{F}_q} \mathbb{F}_q[S]. \tag{50}$$

Equality with K_d is a public rank condition, not a consequence of symmetry alone. Outer randomization does not restore the alternating directions removed before the outer map.

The direct repairs are to increase parameters after accounting for (50), remove public-matrix symmetry, or redesign the left and right power actions so that exchanging their labels changes the quadratic polynomial. The symmetric-block rejection rule is not sufficient: the fixed-skew $\ell = 2$ lift makes every block nonsymmetric by construction, and the structured $\ell = 4$ preflight can certify the same property block by block. Returning unchanged to the submitted $q = 16$ design would instead restore the known characteristic-two attack surface.

A revised security analysis should publish the quotient rank, affine or filter rank, stable-kernel rank when applicable, the polar or cross-polar condition used for target availability, and solver logs for the actual cost-driving system.

10 Conclusion

The central vulnerability is an exact duplicate-feature identity. SNOVA labels a quadratic power-pair feature by an ordered pair (ξ, η), but symmetry of the public base matrices makes that polynomial identical to the one labeled (η, ξ). The public $q = 19$ verifier therefore has at most $m_1\binom{\ell+1}{2}$ distinct power-pair features where the ordered representation has $m_1\ell^2$. Because the identification occurs before the public outer map, no later randomization can restore the missing directions.

The rest of the attack turns that identity into complete forgery systems rather than merely a distinguisher or a reduced subset of verifier equations. Exact affine elimination retains every output. A stable A-linear slice preserves the four-block structure needed for the six $\ell = 4$ special-XL systems. Fixed-skew and target-filtered relations handle the three $\ell = 2$ systems and their block-symmetry rejection rule. Exact moment formulas distinguish root existence, unique-root retries, and the separate question of whether the chosen Macaulay matrix exposes the root. Every candidate is checked by the unmodified verifier.

Under the stated global-rank and production-solver assumptions, the modeled expected operation counts are $2^{128.82}$–$2^{129.44}$, $2^{171.69}$–$2^{183.44}$, and $2^{214.12}$–$2^{235.19}$, and the validated AXN circuit conversion gives $2^{132.17}$–$2^{133.75}$, $2^{175.04}$–$2^{187.74}$, and $2^{217.48}$–$2^{239.50}$—9.25–54.52 bits below NIST's nominal gate references and 9.13–53.86 bits below the released full-key-space AES calibration circuits in the same basis.

The symmetry quotient itself is unconditional: it strictly reduces the quadratic feature space of every symmetric $q = 19$ public key, and every tested full-size instance yields the complete residual system analyzed here. The headline exponents rest on two further inputs that we sample but do not prove—the global polar rank spectra and the production-scale behavior of the MQ solver. Certifying those would turn the reported budgets into a definitive security classification.

For a redesign, the immediate lesson is simple: count distinct quadratic polynomials after every public symmetry, not the labels used to name them.

A Proofs Deferred from the Main Text

The main text states the probability and structural results at the point where an attacker uses them. This appendix collects the longer proofs in the same order.

A.1 Stable-lift kernel

Proof of Theorem 3. Choose an $\mathbb{F}_q$-basis of A. The displayed spanning family in (13) then has $m_1 d^2$ elements, proving the dimension bound. For $\zeta \in A$,

$$(w^{\mathsf{T}}\Sigma_\xi P_i \Sigma_\eta)\Sigma_\zeta = w^{\mathsf{T}}\Sigma_\xi P_i \Sigma_{\eta\zeta},$$

so $\mathcal{U}_w$ is closed under right multiplication by A.

Consider a cross term produced by substituting $u_j = \Sigma_{R_j} u + \Sigma_{\Gamma_j} w$ into a feature. A term with w on the left already has row coefficient $w^{\mathsf{T}}\Sigma_\xi P_i \Sigma_\eta$. A term with w on the right can be transposed; using $S^{\mathsf{T}} = S$, $P_i^{\mathsf{T}} = P_i$, and commutativity of A, its row coefficient has the same form with the two algebra labels reversed. Thus every row of $\mathcal{F}_{R,\mathbf{v}}$ lies in $\mathcal{U}_w$. Consequently every row of $C_{R,\mathbf{v}} = W_R \mathcal{E} \mathcal{F}_{R,\mathbf{v}}$ lies there as well. Right-A stability then puts every row of the orbit matrix $\mathcal{O}_{R,\mathbf{v}}$ in $\mathcal{U}_w$.

If $\operatorname{rank} \mathcal{O}_{R,\mathbf{v}} = m_1 d^2$, then

$$m_1 d^2 = \dim \operatorname{row}(\mathcal{O}_{R,\mathbf{v}}) \le \dim \mathcal{U}_w \le m_1 d^2,$$

so $\operatorname{row}(\mathcal{O}_{R,\mathbf{v}}) = \mathcal{U}_w$. Since the rows of $\mathcal{F}_{R,\mathbf{v}}$ belong to this row space, $\mathcal{F}_{R,\mathbf{v}} \ker \mathcal{O}_{R,\mathbf{v}} = 0$. To see A-stability of the kernel, let $x \in \ker \mathcal{O}_{R,\mathbf{v}}$ and $\zeta \in A$. Every product $S^a\zeta$ is an $\mathbb{F}_q$-linear combination of the chosen basis powers, so $C_{R,\mathbf{v}}\Sigma_{S^a}\Sigma_\zeta x = 0$ for all a; hence $\Sigma_\zeta x \in \ker \mathcal{O}_{R,\mathbf{v}}$. Rank–nullity gives $\dim_{\mathbb{F}_q} H_{R,\mathbf{v}} = N_0 - m_1 d^2$. When $A \cong \mathbb{F}_{q^\ell}$, $H_{R,\mathbf{v}}$ is an A-vector space, and division by ℓ gives $\dim_A H_{R,\mathbf{v}} = n - m_1 \ell$. □

A.2 Stable affine slices

Proof of Corollary 5. The first two ranks in (15) give the full-lift case of Proposition 1; in particular, every target determines a nonempty affine space $a(t) + \ker C_{R,\mathbf{v}}$. The orbit matrix contains $C_{R,\mathbf{v}}$ as its first block, so $H_{R,\mathbf{v}} \subseteq \ker C_{R,\mathbf{v}}$. Hence $a(t) + y + L \subseteq a(t) + \ker C_{R,\mathbf{v}}$ whenever $y \in \ker C_{R,\mathbf{v}}$. Because $L \subseteq H_{R,\mathbf{v}}$, Theorem 3 gives $\mathcal{F}_{R,\mathbf{v}} L = 0$, and therefore $\mathcal{F}_{R,\mathbf{v}} u$ is constant on that slice.

By Lemma 4, the unordered power-pair features can be changed invertibly to eigenblock coordinates. Because Q_R has full column rank, row reduction of the complete output system isolates K rows whose homogeneous quadratic parts are an invertible linear image of z_{sym}. Applying the inverse image change to those rows expresses the quadratic parts in eigenblock coordinates. Full rank of the quadratic feature map restricted to L ensures that this restriction loses no eigenblock component. Translating a diagonal feature introduces only quadratic, linear, and constant terms from one block; translating an off-diagonal feature introduces terms from its two blocks. If h_a is a homogenizer for block a, multiplication by the unique powers of the h_a turns these affine polynomials into exact multidegrees $2\mathbf{e}_a$ and $\mathbf{e}_a + \mathbf{e}_b$.

Distinct unordered multidegrees have disjoint monomial supports. Each component is generated by the m_1 base forms and therefore has rank at most m_1. The total restricted rank is the sum of these component ranks. Since there are $\binom{\ell+1}{2}$ components and the total is $K = m_1\binom{\ell+1}{2}$, every component must have rank exactly m_1. □

A.3 Accepted targets for the $\ell = 2$ filter

Lemma 13 (Quadratic fiber bound)**.** *Let q be odd and let $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$ be quadratic. For each $b \in \mathbb{F}_q^K \setminus \{0\}$, let $r(b)$ be the polar rank of the scalar quadratic $b \cdot g$, put $r_* = \min_{b\neq 0} r(b)$, and set $\alpha = 1 - q^{-K}$. Then every target $z \in \mathbb{F}_q^K$ satisfies*

$$\Big| |g^{-1}(z)| - q^{N-K} \Big| \leq \alpha q^{N-r_*/2}. \tag{51}$$

More generally, if $W \subseteq \mathbb{F}_q^N$ is an affine subspace of codimension c, then

$$|\{x \in W : g(x) = z\}| \leq q^{N-c-K} + \alpha q^{N-r_*/2}. \tag{52}$$

Proof. Fix a nontrivial additive character ψ of $\mathbb{F}_q$. Fourier inversion gives

$$|g^{-1}(z)| = q^{-K} \sum_{b\in\mathbb{F}_q^K} \psi(-b \cdot z) \sum_{x\in\mathbb{F}_q^N} \psi(b \cdot g(x)).$$

The $b = 0$ term is q^{N-K}. For $b \neq 0$, let $r = r(b)$. An invertible linear change of variables splits the polar form into a nondegenerate r-dimensional part and a radical of dimension $N - r$. If the linear part of $b \cdot g$ is nonzero on the radical, summing over the radical gives zero. Otherwise the radical contributes q^{N-r}, while diagonalizing the nondegenerate quadratic part reduces its sum to a product of one-dimensional Gauss sums of absolute value $q^{1/2}$. In either case the inner sum has absolute value at most $q^{N-r/2}$ [9]. Summing the $q^K - 1$ nonzero Fourier coefficients proves (51).

Write $W = a + U$, where U has codimension c. Translation does not change a quadratic polynomial's polar form. Restricting a bilinear form to $U \times U$ lowers its rank by at most $2c$: after extending a basis of U to one of $\mathbb{F}_q^N$, deleting the last c rows and columns changes matrix rank by at most $2c$. Thus every nonzero scalar combination of $g|_W$ has polar rank at least $r_* - 2c$. If $r_* \geq 2c$, the same Fourier calculation on the $(N - c)$-dimensional affine space W bounds each nonzero character sum by

$$q^{(N-c)-(r_*-2c)/2} = q^{N-r_*/2}.$$

If $r_* < 2c$, the trivial bound q^{N-c} is already at most $q^{N-r_*/2}$. This proves (52) in both cases. □

Proof of Proposition 7. By Lemma 13, every target fiber has between $q^{N-K} - \alpha q^{N-r_*/2}$ and $q^{N-K} + \alpha q^{N-r_*/2}$ roots. A rejected root lies on the intersection of some t_{rej} block-symmetry hyperplanes. Each such intersection is an affine subspace of codimension t_{rej}, so (52) bounds the number of roots it contains by $q^{N-t_{\mathrm{rej}}-K} + \alpha q^{N-r_*/2}$. Subtracting the union bound over the B intersections from the full-fiber lower bound gives

$$q^{N-K}(1 - Bq^{-t_{\mathrm{rej}}}) - \alpha q^{N-r_*/2}(1 + B).$$

This is positive exactly under the sufficient inequality (20).

Every fiber has size at most $q^{N-K}(1+\alpha q^{K-r_*/2})$. Since $|\mathcal{X}_{\mathrm{acc}}| = a_n q^N$, double-counting accepted assignments by their images proves (21). Finally, let $T_z = |g^{-1}(z)|$ and let R_z be the number of rejected roots. The accepted fraction is $1 - R_z/T_z$. Applying the union-bound upper estimate to R_z and the character-sum lower estimate to T_z gives (22). □

A.4 Root collisions and the rank-spectrum bound

Proof of Theorem 8. Average over the $q^{\dim V - h}$ cosets and q^K targets. Every point of V contributes once to the first moment, which gives (27). For an ordered pair of distinct points in one coset, write the second point as $x + y$ with $y \in L \setminus \{0\}$. The collision equation is

$$g(x+y) = g(x) \quad \Longleftrightarrow \quad T_y(x) = -(g(y) - g(0)).$$

It has $\kappa(y) q^{\dim V - r(y)}$ solutions. Dividing the total number of ordered collisions by the number of target–coset pairs proves (28).

For the rank-spectrum identity, note that $q^{-r(y)} = q^{-K} |\ker T_y^*|$ and double-count pairs (b, y) satisfying $b \cdot T_y = 0$. The term $b = 0$ contributes $q^h - 1$. For $b \neq 0$, the number of nonzero $y \in L$ annihilated by $b \cdot T_y$ is $q^{h - e_L(b)} - 1$. Therefore

$$\bar{\eta} = q^{-K} \left((q^h - 1) + \sum_{b \neq 0} (q^{h - e_L(b)} - 1) \right) = \mu(1 + \Theta_L(g)) - 1.$$

The inequality $\eta_0 \leq \bar{\eta}$ is immediate.

Using $\mathbb{E}[Z^2] = \mu(1 + \eta_0)$ gives

$$\mathrm{Var}(Z) = \mu(1 + \eta_0 - \mu) \leq \mu^2 \Theta_L(g),$$

and the second-moment inequality proves (31). Finally, $e_* \geq h - \Delta$ implies $\Theta_L(g) \leq (q^K - 1) q^{-h+\Delta} < q^{\tau + \Delta}$. □

A.5 Unique-root trials and the planted distribution

Proof of Corollary 9. For every nonnegative integer z,

$$\mathbf{1}_{z=1} \geq z - z(z-1), \qquad \mathbf{1}_{z>0} \geq z - \binom{z}{2}, \qquad \mathbf{1}_{z\geq 2} \leq \binom{z}{2}.$$

Apply these inequalities to Z and use (28) together with $\eta_0 \leq \bar{\eta}$. Dividing the upper bound on $\Pr[Z \geq 2]$ by (34) proves (35).

The planted law size-biases a target–coset pair by Z. Hence

$$\Pr_{\mathsf{P}}[Z \geq 2] = \frac{\mathbb{E}[Z \mathbf{1}_{Z\geq 2}]}{\mu} \leq \frac{\mathbb{E}[Z(Z-1)]}{\mu} = \eta_0.$$

Conditioned on $Z = 1$, both U and P are uniform on the same set of target–coset pairs. If two distributions have the same conditional law on an event, their total variation distance is at most the larger mass they assign to its complement. Here the complement is $\{Z \geq 2\}$. The same indicator inequality gives the sharper exact-parameter lower bound $\Pr[Z > 0] \geq \mu(1 - \eta_0/2)$. Combining it with $\Pr[Z \geq 2] \leq \mathbb{E}[Z(Z-1)]/2 = \mu\eta_0/2$ gives

$$\Pr_{\mathsf{U}}[Z \geq 2] \leq \frac{\eta_0}{2 - \eta_0} \leq \eta_0 \qquad (\eta_0 < 1).$$

The displayed size-bias bound gives $\Pr_{\mathsf{P}}[Z \geq 2] \leq \eta_0$. Both are at most $\bar{\eta}$, proving the total-variation claim. □

A.6 Full-space target availability

Proof of Corollary 11. Take $L = V$ in Theorem 8. Then $e_L(b)$ is the polar rank and $\Theta_L(g) \le (q^K - 1)q^{-r_*} = \alpha q^{K-r_*}$. For the last statement, (51) gives

$$|g^{-1}(z)| \ge q^{N-K} - \alpha q^{N-r_*/2} > 0$$

when $r_* \ge 2K$. $\square$

B Dimension Formulas

For a Version 2.3 row,

$$n = v + o, \qquad m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \qquad M = or\ell, \qquad N_0 = n\ell, \qquad N_{\text{raw}} = m_1\ell^2 r^2, \qquad K = m_1\binom{\ell+1}{2}.$$

For a full structured lift, $\operatorname{rank} C_{R,\mathbf{v}} = M - K$ and

$$\dim\ker C_{R,\mathbf{v}} = N_0 - M + K.$$

If $\operatorname{rank} \mathcal{O}_{R,\mathbf{v}} = m_1\ell^2$, then

$$\dim_{\mathbb{F}_q} H_{R,\mathbf{v}} = N_0 - m_1\ell^2, \qquad \dim_A H_{R,\mathbf{v}} = n - m_1\ell.$$

The selected special-XL slice has A-dimension s, base-field dimension $h = \ell s$, and retry exponent $\tau = K - h$.

Table 11: Structured-lift dimensions for the six Version 2.3 $\ell = 4$ rows.

Parameters	M	K	$\dim_A H$	s	τ
$(28, 5, 4, 4)$	80	50	13	10	10
$(28, 4, 4, 5)$	80	50	12	10	10
$(40, 7, 4, 4)$	112	70	19	15	10
$(38, 5, 4, 5)$	100	70	15	15	10
$(50, 9, 4, 4)$	144	90	23	19	14
$(52, 6, 4, 6)$	144	90	22	19	14

C Cost-Search Certificate

For each $m_1 \in \{5, 7, 9\}$ and every admissible integer s, the artifact expands (45) exactly. Here admissible means $1 \leq s \leq \min\{\dim_A H, \lfloor K/4 \rfloor\}$. The $K/4$ bounds are $12, 17, 22$. They bind or tie for both $m_1 = 5$ rows and both $m_1 = 9$ rows. For $m_1 = 7$, the bounds are 17 for $(40, 7, 4, 4)$ and 15 for $(38, 5, 4, 5)$, the latter coming from $\dim_A H$. The artifact searches to the larger group bound and then filters the certificate separately for each parameter set; both $m_1 = 7$ rows attain the same optimum at $s = 15$. The monomial count $M_s(\mathbf{d})$ is coordinatewise increasing, so any multidegree that could match or beat the incumbent lies in a finite rectangular box. By symmetry it suffices to enumerate sorted multidegrees. The search checks 3,690, 11,217, and 27,449 candidates for $m_1 = 5, 7, 9$, respectively.

The optima, unique up to block permutation, are those in Table 6. No zero signed coefficient occurs anywhere in these bounded searches, so using a strictly negative rather than a non-positive trigger does not change the result. The optimal coefficients are

$$-14{,}677{,}784, \qquad -124{,}983{,}335{,}190, \qquad -1{,}164{,}044{,}975{,}088{,}360.$$

The corresponding Macaulay column counts are

$$23{,}417{,}049{,}656, \quad 44{,}237{,}557{,}981{,}725{,}696, \quad 236{,}094{,}291{,}515{,}544{,}000{,}000.$$

Substitution of the full-rank retry bound $\mathsf{Retry}_{K,\tau}$ into (47) gives published-model exponents 128.81713, 171.68562, and 214.12301. Replacing b_4 by the validated signed AXN circuit g_4^{AXN} gives converted arithmetic exponents 132.17187, 175.04035, and 217.47775. The NAND sensitivity gives 133.04031, 175.90879, and 218.34618.

D Same-Basis Ordered-Label Estimator Baseline

The comparison with the official Version 2.3 table mixes cost conventions, so we also recost a reproducible *estimator baseline* under exactly the same AXN arithmetic basis as the selected attacks. This baseline feeds the full ordered-label equation/variable dimensions from the earlier affine-column accounting into the pinned generic-MQ optimizer and replaces each charged base-field operation by $g_1^{\mathrm{AXN}} = 441$. An attacker can indeed feed the unreduced affine-column equations to a generic solver. What is counterfactual is the estimator's treatment of the ordered labels as though they represented independent generic quadratic features: on every symmetric key, reversed labels define the same polynomial by Theorem 2. Thus the table is a controlled historical estimator comparison, not evidence for a second quadratic space of the stated ordered-label rank. It reports every row rather than only the range quoted in the introduction.

Table 12: Historical ordered-label generic-MQ estimator and selected attack, both recosted in the same AXN arithmetic basis. "Estimator input" gives the equation/variable dimensions of the unreduced affine-column equations. The system is executable, but treating its reversed labels as independent generic quadratic features is counterfactual on a symmetric key.

Level	Parameters	Estimator input	Ordered-label AXN	Selected AXN	Saving
I	$(28, 5, 4, 4)$	80/132	206.69	132.17	74.52
I	$(48, 16, 2, 2)$	64/128	171.77	133.75	38.02
I	$(28, 4, 4, 5)$	80/128	206.69	132.17	74.52
III	$(40, 7, 4, 4)$	112/188	277.52	175.04	102.48
III	$(72, 24, 2, 2)$	96/192	242.15	187.74	54.41
III	$(38, 5, 4, 5)$	100/172	250.29	175.04	75.25
V	$(50, 9, 4, 4)$	144/236	345.49	217.48	128.01
V	$(96, 32, 2, 2)$	128/256	311.08	239.50	71.58
V	$(52, 6, 4, 6)$	144/232	347.13	217.48	129.66

The exact unrounded savings are 38.02124–129.65538 bits. This comparison still inherits the generic-MQ and special-XL solving models, but it removes the field-operation and gate-basis mismatch from the comparison itself.

E Fixed-Skew Certificate

For the official $\ell = 2$ matrix, direct arithmetic gives

$$9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix}, \qquad e_0^{\mathsf{T}}(9I + 10S) = e_1^{\mathsf{T}}.$$

Together with (17), this proves (18) without a rank or genericity assumption. The remaining public preflight checks $\operatorname{rank} Q_R = K$ and $\operatorname{rank} C_{R,\mathbf{v}} = M - K$.

The resulting residual dimensions are 48/112, 72/168, and 96/224. The reoptimized generic-MQ convention gives exponents 129.44030, 183.43803, and 237.84310. The last optimum uses Hashimoto's $a = 1$ boundary and the direct MQ(19, 1, 1) solver described in the main text. Replacing the coarse factor b_1 by the validated signed AXN circuit g_1^{AXN} gives converted arithmetic exponents 133.74643, 187.74416, and 242.14922. By Corollary 11, a global polar-rank lower bound r adds at most

$$\log_2\Big(1 + (1 - 19^{-K})19^{K-r}\Big)$$

bits for target retries. Requiring positive AXN arithmetic budget below the nominal NIST references gives thresholds 46, 68, and 89. The stronger condition $r \geq 2K$, namely 96, 144, and 192, makes every target solvable. These are the values in Table 7 and Table 10.

F Artifact and Reproduction

The release contains the complete paper source, bibliography, and a self-contained evidence directory with:

- the public-key generator, direct verifier, and official Level-I known-answer public key and signature;
- the quotient, affine-feature, structured-offset, stable-kernel, and rejection checks;

- all 121 structured preflight records, all 13 structured-slice records, and 60 fixed-skew preflights;
- 102,796 structured cross-polar ranks, 21,992 dual directional-derivative ranks, 3,888 fixed-skew polar ranks, and 29,424 zero-offset filter ranks, including the exceptional-direction campaigns;
- exhaustive $\ell = 2$ relation enumeration, both rejection certificates, accepted-target bounds, and generic-MQ costs;
- the complete special-XL and two-block cost searches, 128 exact $\ell = 4$ and 48 control Macaulay experiments, and the target-generation bound;
- 110 exhaustive toy checks of the exact collision and rank-spectrum formulas, including rank-defective and compatibility-defective cases, plus genericity diagnostics on six actual $\ell = 2$ headline cores;
- an independent rank implementation checked on 2,036 matrices, a projective-distinctness audit of all 124,788 structured rank-test directions, and independent scalar evaluators for the released gate circuits and headline cost conversions.

All finite-field ranks are computed exactly over $\mathbb{F}_{19}$. The release records hashes of the scripts, public keys, stable kernels, selected slices, and polar bases, together with a manifest for every file.

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